

Membership-inference audit report

Generated 2026-08-13 20:30

22 run(s) collected

run directory	type	algorithm	dataset*	roles	curves

sweep_l2_cifar100_gaussian_noise	level2	gaussian_noise	cifar100	3	3
sweep_l2_cifar100_neggrad	level2	neggrad	cifar100	3	3
sweep_l2_cifar100_neggrad_plus	level2	neggrad_plus	cifar100	3	3
sweep_l2_cifar100_random_label	level2	random_label	cifar100	3	3
sweep_l2_cifar100_wig	level2	wig	cifar100	3	3
sweep_l2_cifar10_gaussian_noise	level2	gaussian_noise	cifar10	3	3
sweep_l2_cifar10_neggrad	level2	neggrad	cifar10	3	3
sweep_l2_cifar10_neggrad_plus	level2	neggrad_plus	cifar10	3	3
sweep_l2_cifar10_random_label	level2	random_label	cifar10	3	3
sweep_l2_cifar10_wig	level2	wig	cifar10	3	3
sweep_l3_cifar100_gaussian_noise	level3	gaussian_noise	cifar100	3	18
sweep_l3_cifar100_neggrad	level3	neggrad	cifar100	3	18
sweep_l3_cifar100_neggrad_plus	level3	neggrad_plus	cifar100	3	18
sweep_l3_cifar100_random_label	level3	random_label	cifar100	3	18
sweep_l3_cifar100_wig	level3	wig	cifar100	3	18
sweep_l3_cifar10_gaussian_noise	level3	gaussian_noise	cifar10	3	18
sweep_l3_cifar10_neggrad	level3	neggrad	cifar10	3	18
sweep_l3_cifar10_neggrad_plus	level3	neggrad_plus	cifar10	3	18
sweep_l3_cifar10_random_label	level3	random_label	cifar10	3	18
sweep_l3_cifar10_wig	level3	wig	cifar10	3	18
sweep_ola_cifar10	ola	ola p_gen=0.5 p_mem=0.2	cifar10	3	36
sweep_ola_cifar100	ola	ola p_gen=0.5 p_mem=0.2	cifar100	3	36

* dataset is read from the run directory name, not the summary (the summaries do not record it).

Level-2 U-RMIA (naive offline attack)

10 run(s)

Comparison across runs

One row per run. 'unl' is the unlearned model, 'retr' the retrained one. Two-sided AUC is max(AUC, 1-AUC).

run	orig AUC	unl AUC	unl 2-sid	retr AUC	unl test	unl forget	retr forget
cifar100_gaussian_noise	0.7703	0.7240	0.7240	0.4928	0.2520	0.5400	0.2820
cifar100_neggrad	0.7685	0.5206	0.5206	0.5038	0.1060	0.1440	0.3200
cifar100_neggrad_plus	0.7732	0.3495	0.6505	0.4810	0.2960	0.1860	0.2940
cifar100_random_label	0.7853	0.6700	0.6700	0.5086	0.1460	0.2320	0.3020
cifar100_wig	0.7460	0.4911	0.5089	0.5011	0.2660	0.2800	0.2780
cifar10_gaussian_noise	0.7943	0.6511	0.6511	0.5030	0.5800	0.7460	0.6840
cifar10_neggrad	0.7822	0.7871	0.7871	0.5294	0.7200	0.9980	0.7180
cifar10_neggrad_plus	0.7823	0.6274	0.6274	0.4964	0.7260	0.8200	0.7000
cifar10_random_label	0.7812	0.5235	0.5235	0.5122	0.2680	0.2520	0.6840
cifar10_wig	0.7857	0.4979	0.5021	0.5228	0.5820	0.5700	0.6940

sweep_l2_cifar100_gaussian_noise

Level-2 U-RMIA (naive offline attack) · dataset cifar100

Configuration

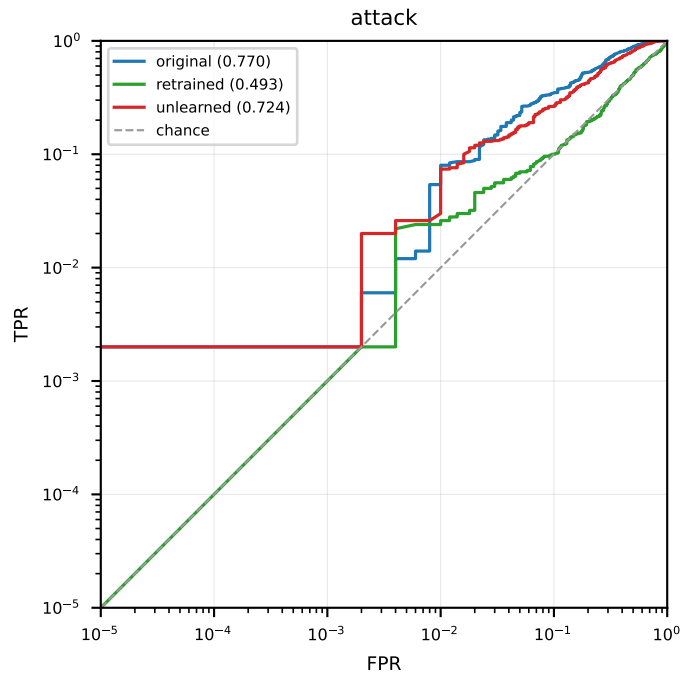
algorithm : gaussian_noise
forget_size : 500
params : noise_std=0.002

role	AUC	2-sided	TPR@1%	TPR@.1%	TPR@0%	test_acc	forget_acc
original	0.7703	0.7703	0.0800	0.0020	0.0020	0.2860	n/a
unlearned	0.7240	0.7240	0.0740	0.0020	0.0020	0.2520	0.5400
retrained	0.4928	0.5072	0.0260	0.0000	0.0000	0.2880	0.2820

Timing

Loading dataset took 2170.57971 seconds
Model loading/training took 4171.5 seconds
Preparing signals took 4.39520 seconds
Auditing took 7.5 seconds
Total runtime: 6354.67544 seconds

sweep_l2_cifar100_gaussian_noise — log-scale ROC



sweep_l2_cifar100_neggrad

Level-2 U-RMIA (naive offline attack) · dataset cifar100

Configuration

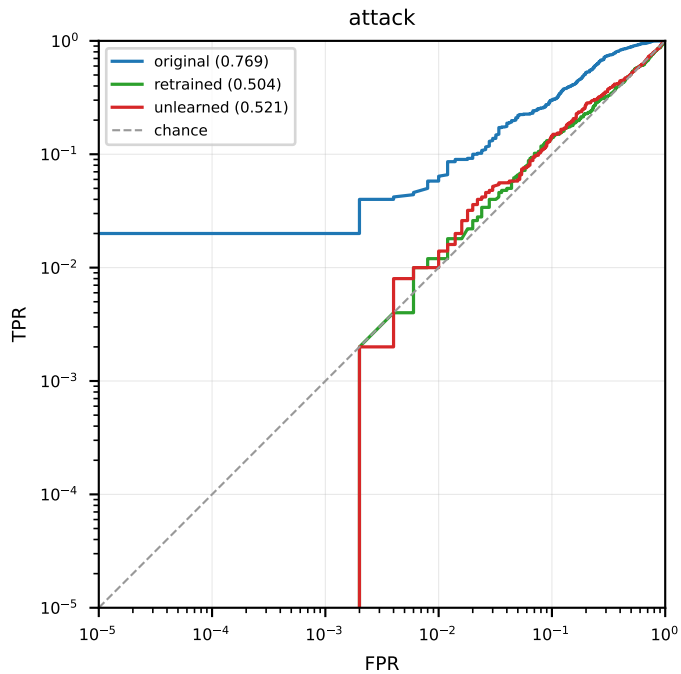
algorithm : neggrad
forget_size : 500
params : epochs=3, learning_rate=0.001

role	AUC	2-sided	TPR@1%	TPR@.1%	TPR@0%	test_acc	forget_acc
original	0.7685	0.7685	0.0640	0.0200	0.0200	0.2520	n/a
unlearned	0.5206	0.5206	0.0140	0.0000	0.0000	0.1060	0.1440
retrained	0.5038	0.5038	0.0120	0.0000	0.0000	0.3080	0.3200

Timing

Loading dataset took 0.16908 seconds
Model loading/training took 3906.1 seconds
Preparing signals took 4.75112 seconds
Auditing took 7.3 seconds
Total runtime: 3918.51157 seconds

sweep_l2_cifar100_neggrad — log-scale ROC



sweep_l2_cifar100_neggrad_plus

Level-2 U-RMIA (naive offline attack) · dataset cifar100

Configuration

algorithm : neggrad_plus

forget_size : 500

params : epochs=2, learning_rate=0.002, forget_coef=0.5, retain_coef=1.0

role	AUC	2-sided	TPR@1%	TPR@.1%	TPR@0%	test_acc	forget_acc
original	0.7732	0.7732	0.0480	0.0180	0.0180	0.2920	n/a
unlearned	0.3495	0.6505	0.0120	0.0020	0.0020	0.2960	0.1860
retrained	0.4810	0.5190	0.0080	0.0000	0.0000	0.3060	0.2940

Timing

Loading dataset took 0.16142 seconds

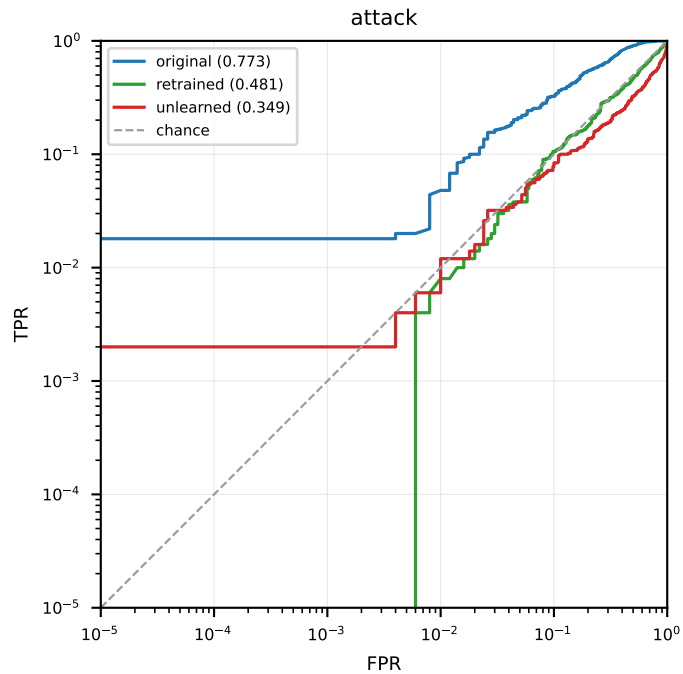
Model loading/training took 3940.0 seconds

Preparing signals took 4.64388 seconds

Auditing took 8.0 seconds

Total runtime: 3952.99497 seconds

sweep_l2_cifar100_neggrad_plus — log-scale ROC



sweep_l2_cifar100_random_label

Level-2 U-RMIA (naive offline attack) · dataset cifar100

Configuration

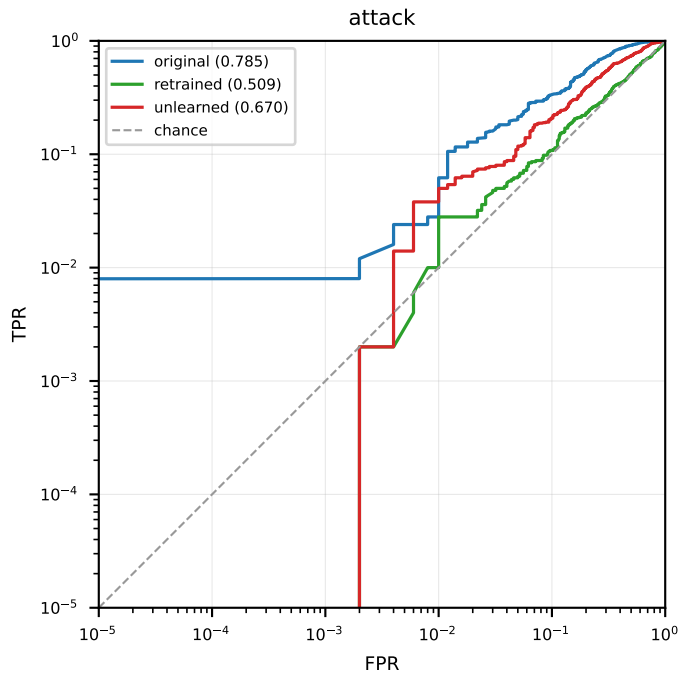
algorithm : random_label
forget_size : 500
params : epochs=3, learning_rate=0.001

role	AUC	2-sided	TPR@1%	TPR@.1%	TPR@0%	test_acc	forget_acc
original	0.7853	0.7853	0.0620	0.0020	0.0020	0.2660	n/a
unlearned	0.6700	0.6700	0.0500	0.0000	0.0000	0.1460	0.2320
retrained	0.5086	0.5086	0.0280	0.0000	0.0000	0.2910	0.3020

Timing

Loading dataset took 0.16890 seconds
Model loading/training took 3908.4 seconds
Preparing signals took 4.58839 seconds
Auditing took 7.9 seconds
Total runtime: 3921.34584 seconds

sweep_l2_cifar100_random_label — log-scale ROC



sweep_l2_cifar100_wig

Level-2 U-RMIA (naive offline attack) · dataset cifar100

Configuration

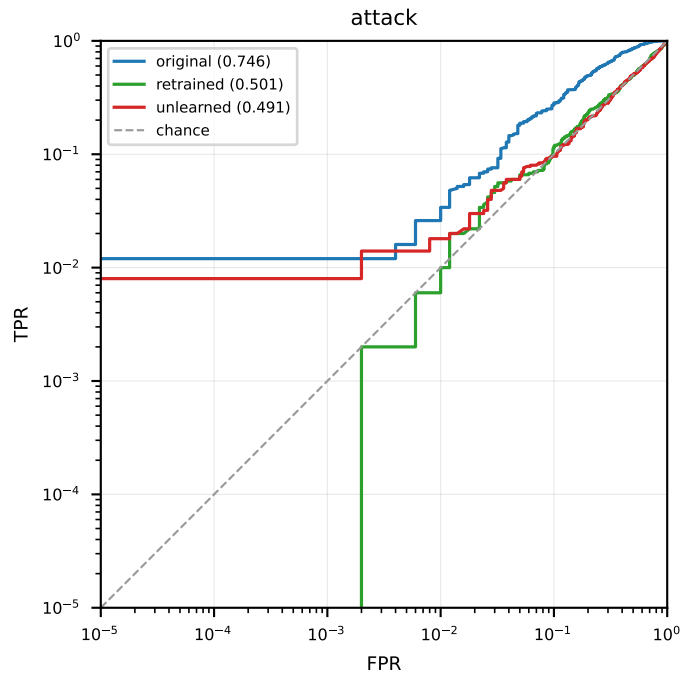
algorithm : wig
forget_size : 500
params : init_ratio=0.3, grad_batch_size=32, epochs=30, learning_rate=0.02

role	AUC	2-sided	TPR@1%	TPR@.1%	TPR@0%	test_acc	forget_acc
original	0.7460	0.7460	0.0340	0.0120	0.0120	0.2840	n/a
unlearned	0.4911	0.5089	0.0180	0.0080	0.0080	0.2660	0.2800
retrained	0.5011	0.5011	0.0100	0.0000	0.0000	0.2820	0.2780

Timing

Loading dataset took 0.11456 seconds
Model loading/training took 3978.0 seconds
Preparing signals took 4.42662 seconds
Auditing took 7.4 seconds
Total runtime: 3990.37493 seconds

sweep_l2_cifar100_wig — log-scale ROC



sweep_l2_cifar10_gaussian_noise

Level-2 U-RMIA (naive offline attack) · dataset cifar10

Configuration

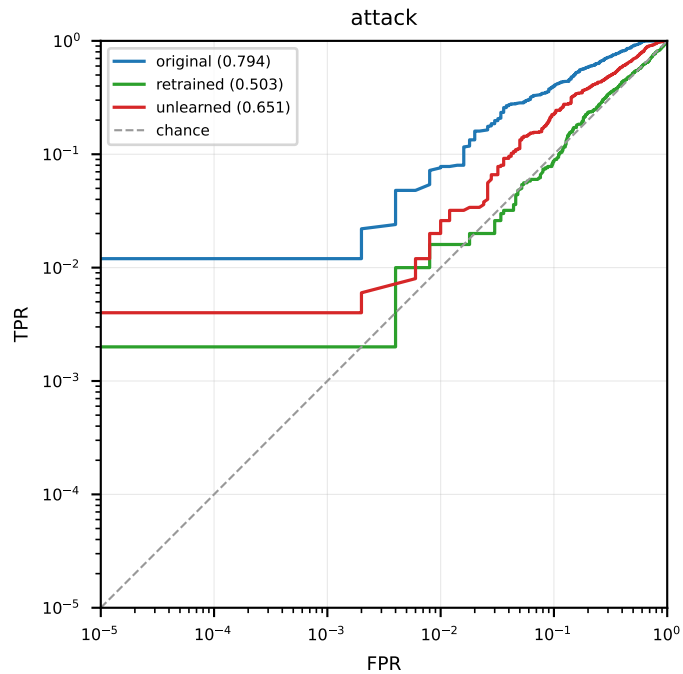
algorithm : gaussian_noise
forget_size : 500
params : noise_std=0.003

role	AUC	2-sided	TPR@1%	TPR@.1%	TPR@0%	test_acc	forget_acc
original	0.7943	0.7943	0.0780	0.0100	0.0100	0.6800	n/a
unlearned	0.6511	0.6511	0.0260	0.0040	0.0040	0.5800	0.7460
retrained	0.5030	0.5030	0.0160	0.0020	0.0020	0.6930	0.6840

Timing

Loading dataset took 2.61223 seconds
Model loading/training took 3922.2 seconds
Preparing signals took 4.40821 seconds
Auditing took 9.3 seconds
Total runtime: 3938.73407 seconds

sweep_l2_cifar10_gaussian_noise — log-scale ROC



sweep_l2_cifar10_neggrad

Level-2 U-RMIA (naive offline attack) · dataset cifar10

Configuration

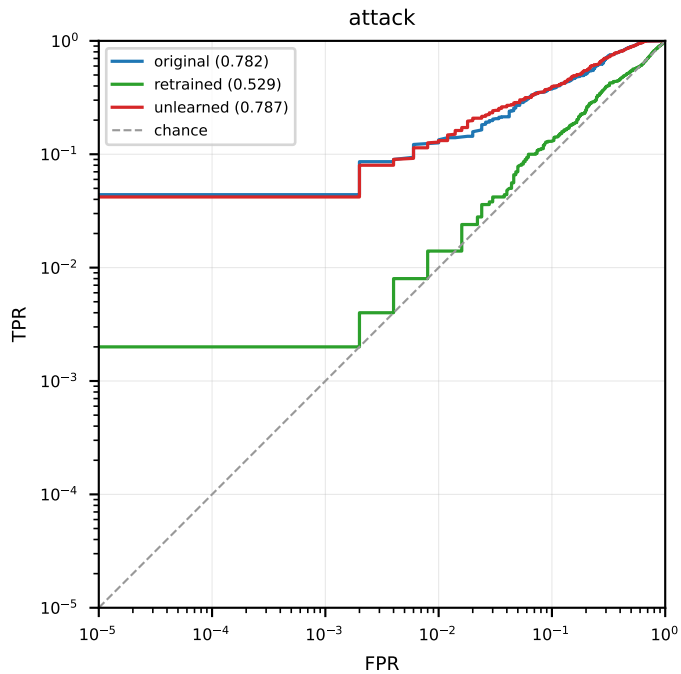
algorithm : neggrad
forget_size : 500
params : epochs=3, learning_rate=0.001

role	AUC	2-sided	TPR@1%	TPR@.1%	TPR@0%	test_acc	forget_acc
original	0.7822	0.7822	0.1340	0.0440	0.0440	0.7120	n/a
unlearned	0.7871	0.7871	0.1320	0.0420	0.0420	0.7200	0.9980
retrained	0.5294	0.5294	0.0140	0.0020	0.0020	0.7170	0.7180

Timing

Loading dataset took 0.09751 seconds
Model loading/training took 3905.1 seconds
Preparing signals took 4.16279 seconds
Auditing took 7.5 seconds
Total runtime: 3917.19624 seconds

sweep_l2_cifar10_neggrad — log-scale ROC



sweep_l2_cifar10_neggrad_plus

Level-2 U-RMIA (naive offline attack) · dataset cifar10

Configuration

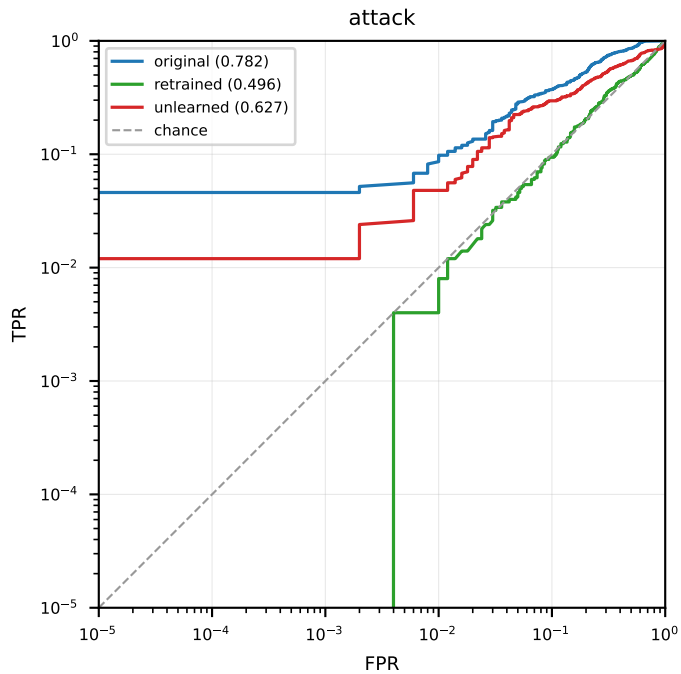
algorithm : neggrad_plus
forget_size : 500
params : epochs=2, learning_rate=0.002, forget_coef=0.5, retain_coef=1.0, forget_loss_cap=1.16

role	AUC	2-sided	TPR@1%	TPR@.1%	TPR@0%	test_acc	forget_acc
original	0.7823	0.7823	0.0980	0.0460	0.0460	0.7120	n/a
unlearned	0.6274	0.6274	0.0480	0.0120	0.0120	0.7260	0.8200
retrained	0.4964	0.5036	0.0080	0.0000	0.0000	0.7170	0.7000

Timing

Loading dataset took 0.09393 seconds
Model loading/training took 3913.6 seconds
Preparing signals took 4.87700 seconds
Auditing took 8.5 seconds
Total runtime: 3927.55813 seconds

sweep_l2_cifar10_neggrad_plus — log-scale ROC



sweep_l2_cifar10_random_label

Level-2 U-RMIA (naive offline attack) · dataset cifar10

Configuration

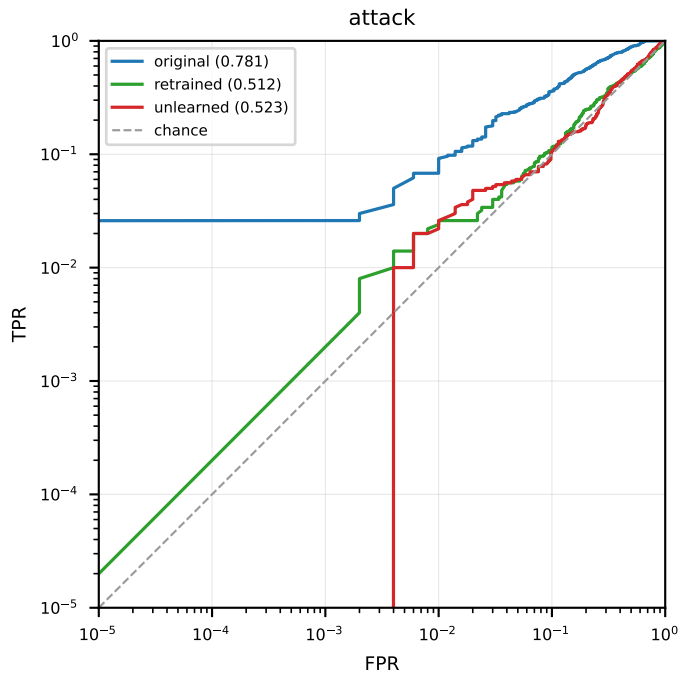
algorithm : random_label
forget_size : 500
params : epochs=3, learning_rate=0.001

role	AUC	2-sided	TPR@1%	TPR@.1%	TPR@0%	test_acc	forget_acc
original	0.7812	0.7812	0.0920	0.0100	0.0100	0.7200	n/a
unlearned	0.5235	0.5235	0.0260	0.0000	0.0000	0.2680	0.2520
retrained	0.5122	0.5122	0.0260	0.0000	0.0000	0.6920	0.6840

Timing

Loading dataset took 0.10999 seconds
Model loading/training took 3931.8 seconds
Preparing signals took 4.78980 seconds
Auditing took 8.9 seconds
Total runtime: 3945.93746 seconds

sweep_l2_cifar10_random_label — log-scale ROC



sweep_l2_cifar10_wig

Level-2 U-RMIA (naive offline attack) · dataset cifar10

Configuration

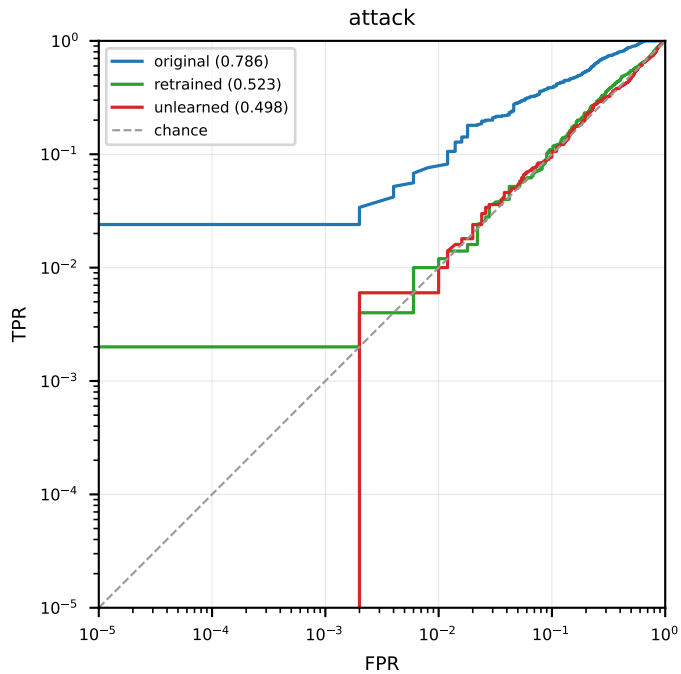
algorithm : wig
forget_size : 500
params : init_ratio=0.3, grad_batch_size=32, epochs=30, learning_rate=0.02

role	AUC	2-sided	TPR@1%	TPR@.1%	TPR@0%	test_acc	forget_acc
original	0.7857	0.7857	0.0760	0.0240	0.0240	0.7280	n/a
unlearned	0.4979	0.5021	0.0100	0.0000	0.0000	0.5820	0.5700
retrained	0.5228	0.5228	0.0120	0.0020	0.0020	0.6950	0.6940

Timing

Loading dataset took 0.18754 seconds
Model loading/training took 3990.8 seconds
Preparing signals took 3.90932 seconds
Auditing took 8.0 seconds
Total runtime: 4003.62022 seconds

sweep_l2_cifar10_wig — log-scale ROC



Level-3 U-RMIA (strong attack, train-then-unlearn references)

10 run(s)

Comparison across runs

Two-sided AUC on the unlearned model, one column per scorer.

run	conf_global	mentr_global	ulira	ulira_mentr	urmia_offline	urmia_online	unl test	unl forget
cifar100_gaussian_noise	0.7181	0.7198	0.5823	0.5964	0.7328	0.7625	0.2340	0.5420
cifar100_neggrad	0.5225	0.5212	0.5242	0.5118	0.5226	0.5230	0.1580	0.1460
cifar100_neggrad_plus	0.6527	0.6538	0.6318	0.6088	0.6570	0.6840	0.2700	0.1840
cifar100_random_label	0.6472	0.6516	0.6109	0.6108	0.6493	0.6553	0.1360	0.2560
cifar100_wig	0.5191	0.5212	0.5127	0.5039	0.5197	0.5185	0.2300	0.2680
cifar10_gaussian_noise	0.5972	0.5924	0.6149	0.6119	0.6377	0.6309	0.5920	0.7400
cifar10_neggrad	0.7021	0.7026	0.7624	0.7969	0.7751	0.7880	0.7040	1.0000
cifar10_neggrad_plus	0.5464	0.5455	0.6485	0.6558	0.6163	0.6150	0.7120	0.8200
cifar10_random_label	0.5039	0.5062	0.5635	0.5386	0.5070	0.5109	0.2880	0.2440
cifar10_wig	0.5208	0.5219	0.5202	0.5104	0.5021	0.5046	0.6380	0.6140

sweep_l3_cifar100_gaussian_noise

Level-3 U-RMIA (strong attack, train-then-unlearn references) · dataset cifar100

Configuration

algorithm : gaussian_noise
forget_size : 500
params : noise_std=0.002

Accuracy

role	test_acc	forget_acc
original	0.2800	n/a
unlearned	0.2340	0.5420
retrained	0.2710	0.2900

AUC

role	conf_global	mentr_global	ulira	ulira_mentr	urmia_offline	urmia_online
original	0.7572	0.7626	0.5733	0.6041	0.7776	0.8185
unlearned	0.7181	0.7198	0.5823	0.5964	0.7328	0.7625
retrained	0.5041	0.5053	0.4749	0.4749	0.5071	0.5014

Two-sided AUC

role	conf_global	mentr_global	ulira	ulira_mentr	urmia_offline	urmia_online
original	0.7572	0.7626	0.5733	0.6041	0.7776	0.8185
unlearned	0.7181	0.7198	0.5823	0.5964	0.7328	0.7625
retrained	0.5041	0.5053	0.5251	0.5251	0.5071	0.5014

TPR @ 0.1% FPR

role	conf_global	mentr_global	ulira	ulira_mentr	urmia_offline	urmia_online
original	0.0020	0.0020	0.0000	0.0060	0.0040	0.0040
unlearned	0.0000	0.0000	0.0000	0.0020	0.0020	0.0280
retrained	0.0020	0.0020	0.0020	0.0020	0.0020	0.0080

TPR @ 1% FPR

role	conf_global	mentr_global	ulira	ulira_mentr	urmia_offline	urmia_online
original	0.0100	0.0100	0.0300	0.0260	0.0440	0.1240
unlearned	0.0280	0.0240	0.0100	0.0380	0.0520	0.1060
retrained	0.0060	0.0060	0.0100	0.0080	0.0120	0.0280

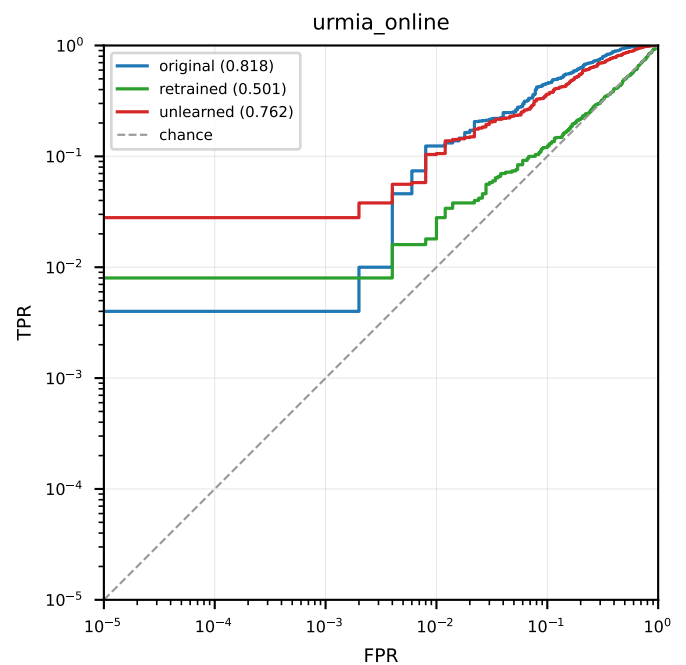
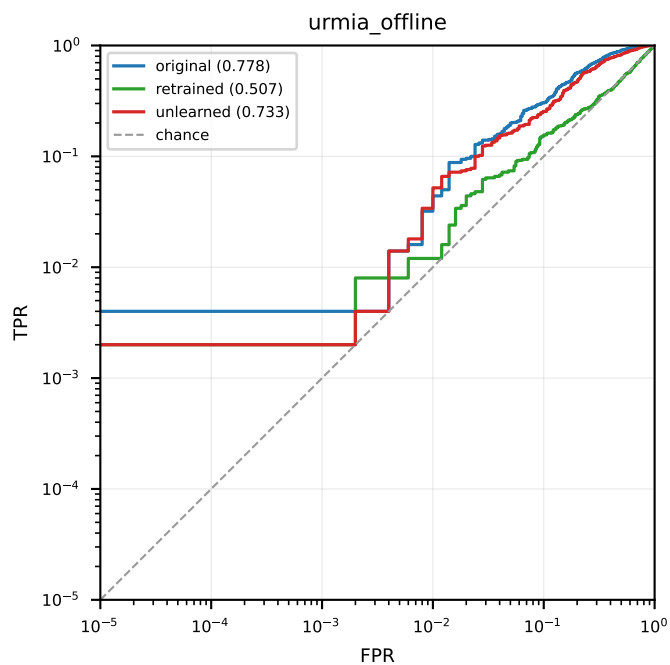
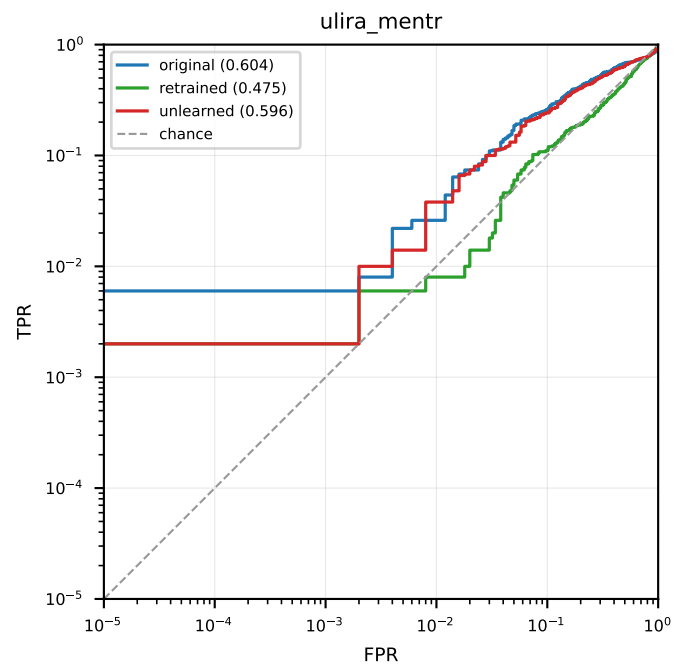
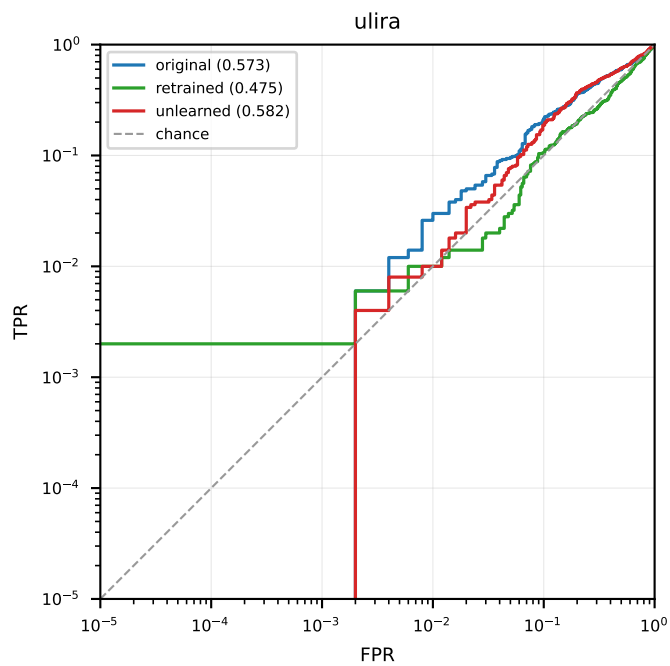
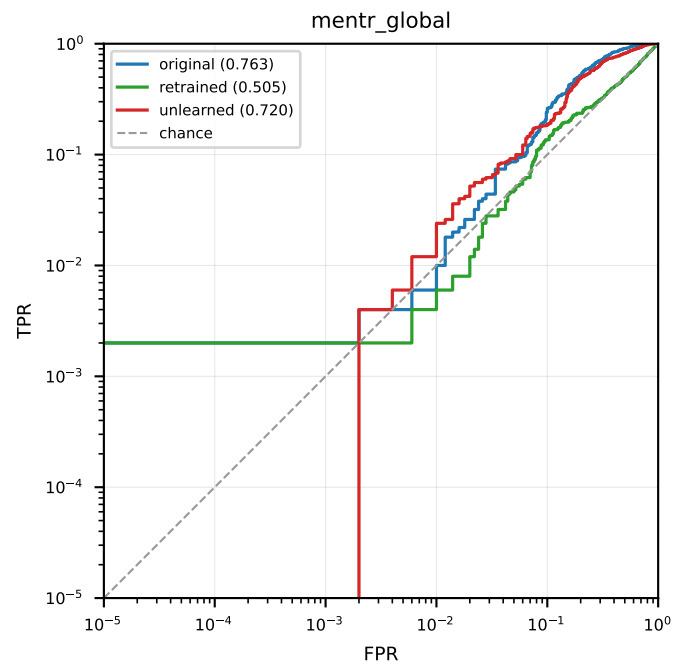
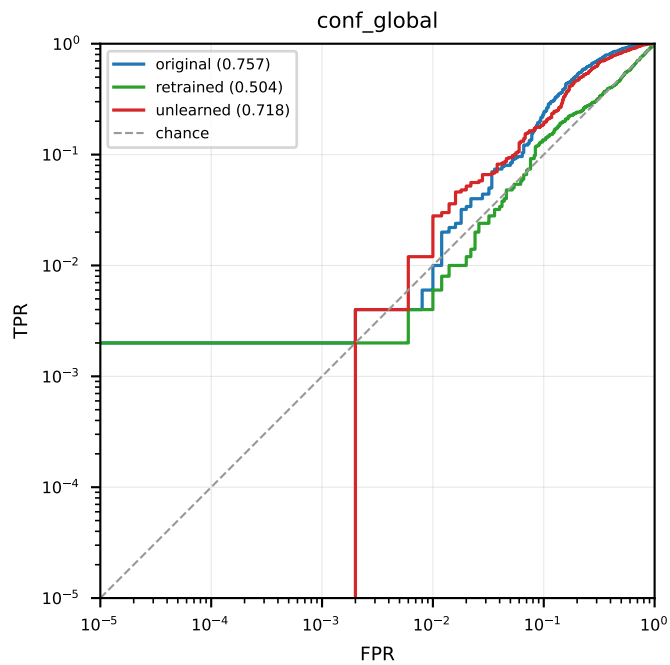
Positive control

dead_scorers : none
null_se : 0.01827
dead_threshold : 0.53653
elevated_threshold : 0.57307
original (two-sided) AUC: urmia_offline=0.7776, urmia_online=0.8185, ulira=0.5733, conf_global=0.7572, mentr_global=0.7626, ulira_mentr=0.6041

Timing

Loading dataset took 0.34737 seconds
Model loading/training took 3921.6 seconds
Preparing signals took 4.94246 seconds
Auditing took 45.6 seconds
Total runtime: 3972.65092 seconds

sweep_I3_cifar100_gaussian_noise — log-scale ROC



sweep_l3_cifar100_neggrad

Level-3 U-RMIA (strong attack, train-then-unlearn references) · dataset cifar100

Configuration

algorithm : neggrad
forget_size : 500
params : epochs=3, learning_rate=0.001

Accuracy

role	test_acc	forget_acc
original	0.2540	n/a
unlearned	0.1580	0.1460
retrained	0.2800	0.2860

AUC

role	conf_global	mentr_global	ulira	ulira_mentr	urmia_offline	urmia_online
original	0.7527	0.7564	0.5219	0.5225	0.7600	0.8003
unlearned	0.5225	0.5212	0.4758	0.4882	0.5226	0.5230
retrained	0.5061	0.5094	0.4822	0.4829	0.5059	0.5048

Two-sided AUC

role	conf_global	mentr_global	ulira	ulira_mentr	urmia_offline	urmia_online
original	0.7527	0.7564	0.5219	0.5225	0.7600	0.8003
unlearned	0.5225	0.5212	0.5242	0.5118	0.5226	0.5230
retrained	0.5061	0.5094	0.5178	0.5171	0.5059	0.5048

TPR @ 0.1% FPR

role	conf_global	mentr_global	ulira	ulira_mentr	urmia_offline	urmia_online
original	0.0000	0.0000	0.0140	0.0020	0.0000	0.0320
unlearned	0.0000	0.0000	0.0000	0.0000	0.0000	0.0080
retrained	0.0080	0.0080	0.0020	0.0020	0.0020	0.0020

TPR @ 1% FPR

role	conf_global	mentr_global	ulira	ulira_mentr	urmia_offline	urmia_online
original	0.0120	0.0120	0.0420	0.0300	0.0240	0.0700
unlearned	0.0100	0.0100	0.0080	0.0060	0.0160	0.0100
retrained	0.0120	0.0120	0.0100	0.0120	0.0040	0.0100

Positive control

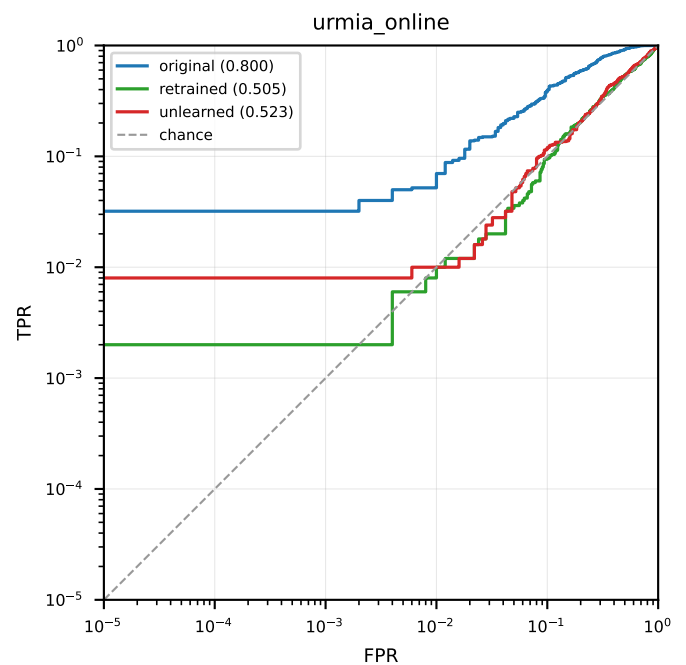
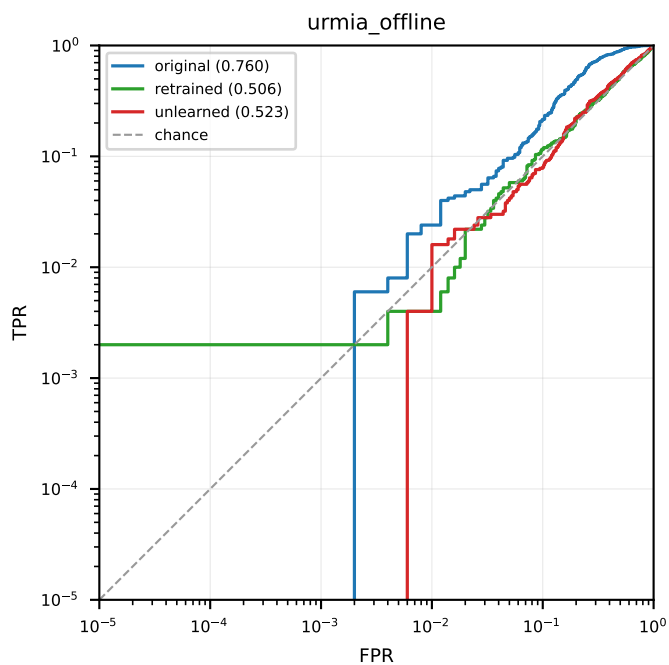
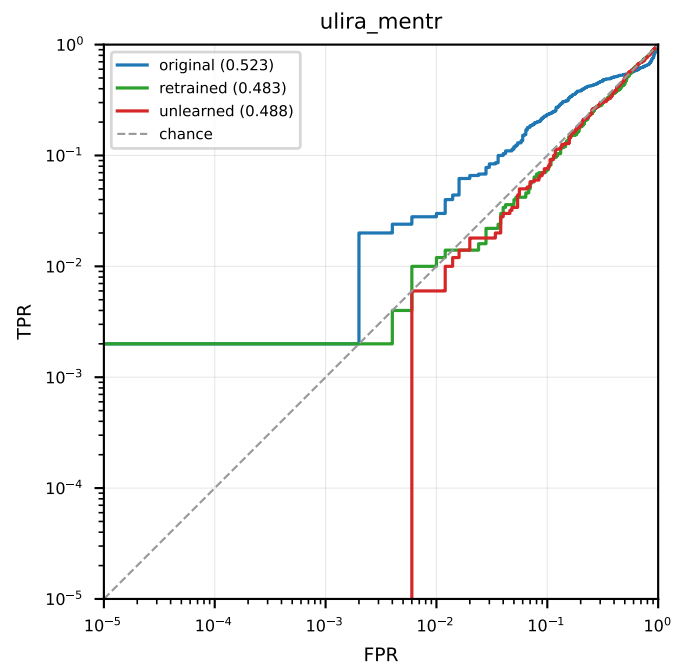
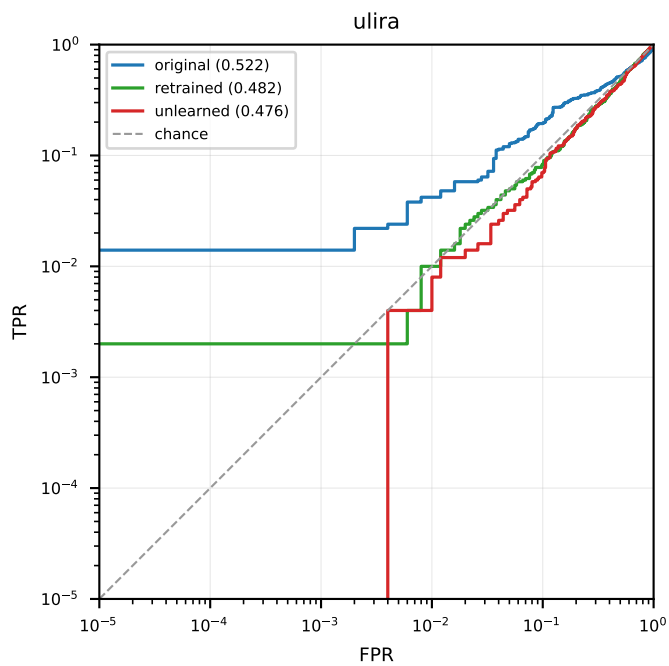
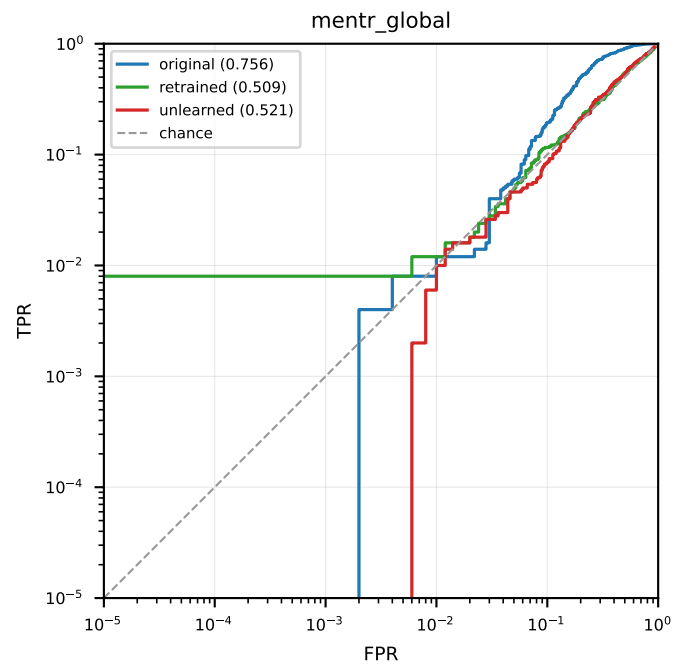
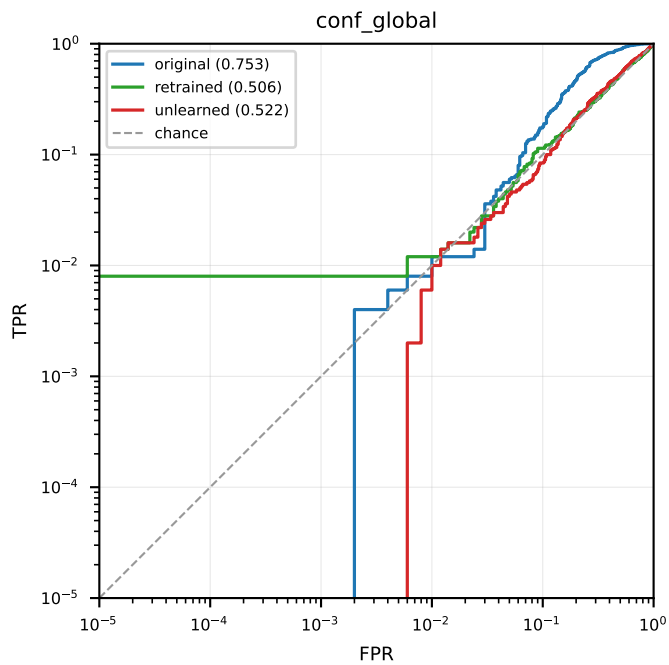
dead_scorers : ['ulira', 'ulira_mentr']
null_se : 0.01827
dead_threshold : 0.53653
elevated_threshold : 0.57307

original (two-sided) AUC: urmia_offline=0.7600, urmia_online=0.8003, ulira=0.5219, conf_global=0.7527, mentr_global=0.7564, ulira_mentr=0.5225

Timing

Loading dataset took 0.09751 seconds
Model loading/training took 3920.7 seconds
Preparing signals took 4.05963 seconds
Auditing took 41.0 seconds
Total runtime: 3965.95048 seconds

sweep_l3_cifar100_neggrad — log-scale ROC



sweep_l3_cifar100_neggrad_plus

Level-3 U-RMIA (strong attack, train-then-unlearn references) · dataset cifar100

Configuration

algorithm : neggrad_plus
forget_size : 500
params : epochs=2, learning_rate=0.002, forget_coef=0.5, retain_coef=1.0

Accuracy

role	test_acc	forget_acc
original	0.2660	n/a
unlearned	0.2700	0.1840
retrained	0.2770	0.2740

AUC

role	conf_global	mentr_global	ulira	ulira_mentr	urmia_offline	urmia_online
original	0.7586	0.7635	0.5255	0.4715	0.7764	0.8069
unlearned	0.3473	0.3462	0.6318	0.6088	0.3430	0.3160
retrained	0.4839	0.4824	0.5298	0.5265	0.4845	0.4713

Two-sided AUC

role	conf_global	mentr_global	ulira	ulira_mentr	urmia_offline	urmia_online
original	0.7586	0.7635	0.5255	0.5285	0.7764	0.8069
unlearned	0.6527	0.6538	0.6318	0.6088	0.6570	0.6840
retrained	0.5161	0.5176	0.5298	0.5265	0.5155	0.5287

TPR @ 0.1% FPR

role	conf_global	mentr_global	ulira	ulira_mentr	urmia_offline	urmia_online
original	0.0020	0.0020	0.0020	0.0060	0.0180	0.0480
unlearned	0.0020	0.0020	0.0140	0.0040	0.0080	0.0000
retrained	0.0020	0.0020	0.0160	0.0020	0.0000	0.0040

TPR @ 1% FPR

role	conf_global	mentr_global	ulira	ulira_mentr	urmia_offline	urmia_online
original	0.0180	0.0160	0.0360	0.1000	0.0780	0.1120
unlearned	0.0140	0.0140	0.0700	0.0200	0.0140	0.0040
retrained	0.0160	0.0160	0.0220	0.0120	0.0100	0.0180

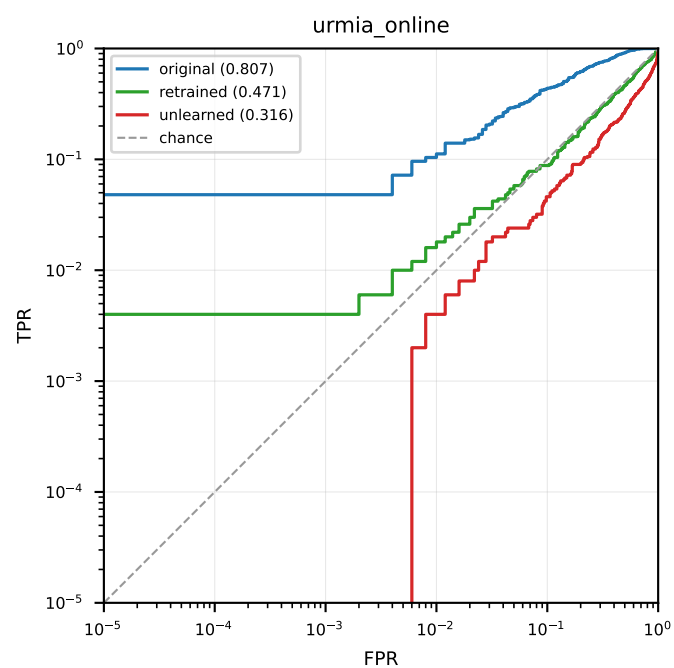
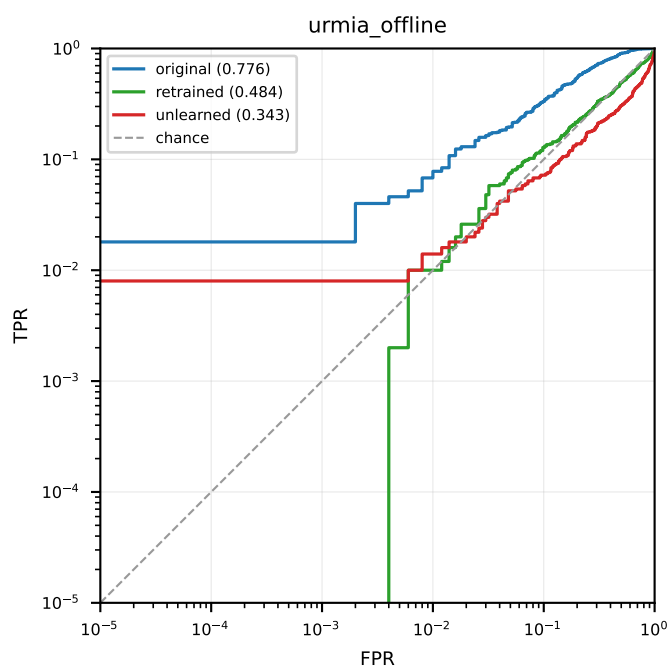
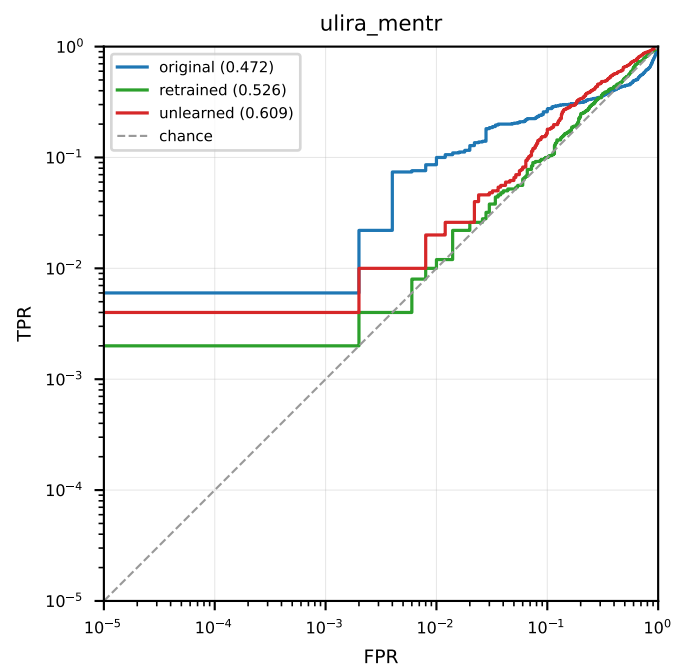
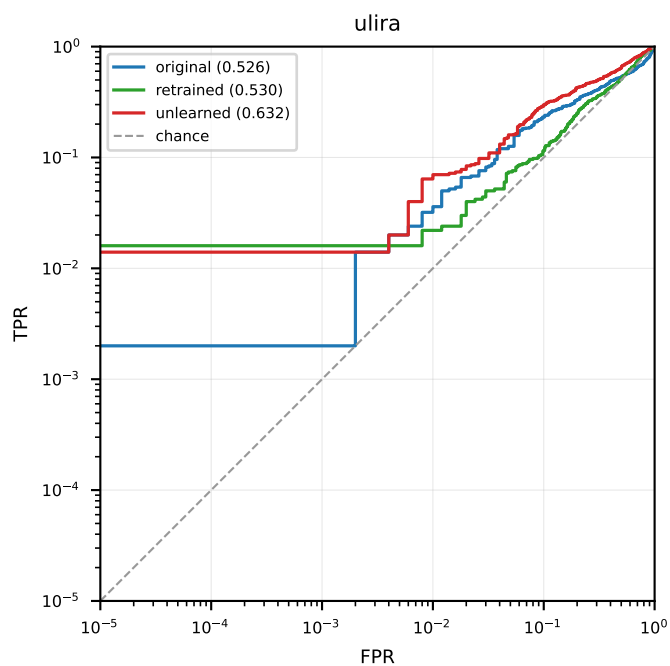
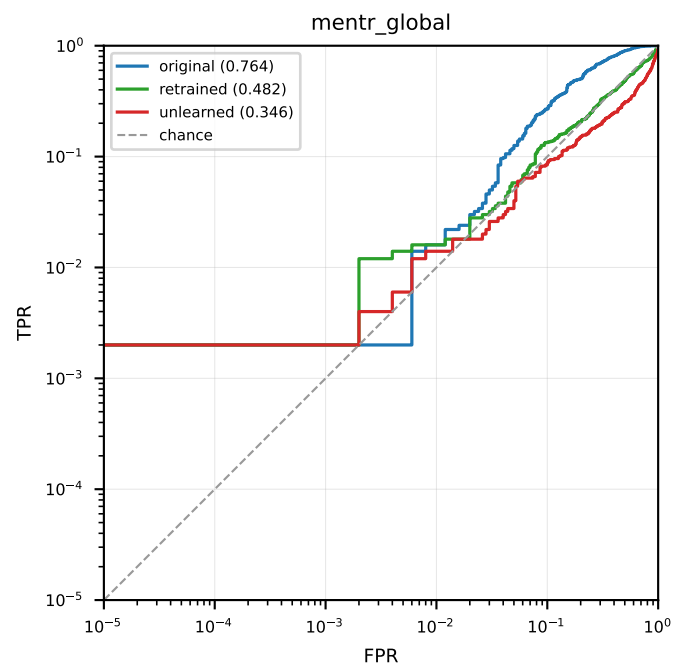
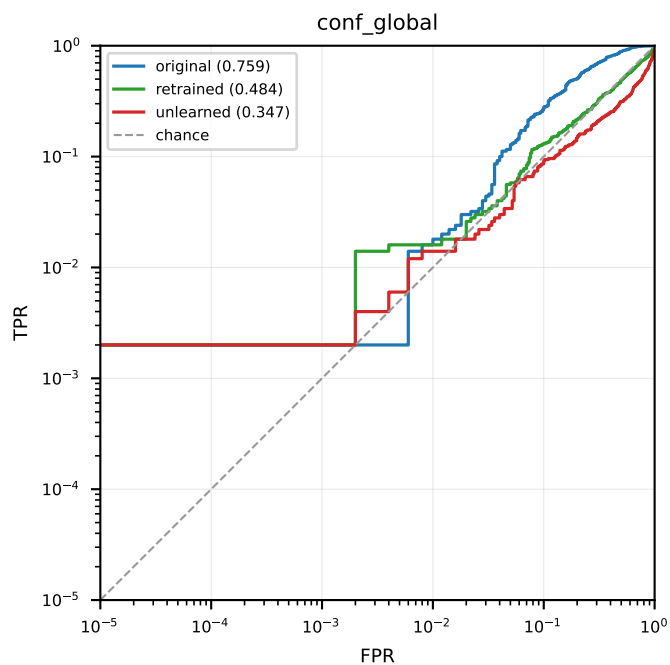
Positive control

dead_scorers : ['ulira', 'ulira_mentr']
null_se : 0.01827
dead_threshold : 0.53653
elevated_threshold : 0.57307
original (two-sided) AUC: urmia_offline=0.7764, urmia_online=0.8069, ulira=0.5255, conf_global=0.7586, zentr_global=0.7635, ulira_mentr=0.5285

Timing

Loading dataset took 0.09486 seconds
Model loading/training took 4072.9 seconds
Preparing signals took 4.42560 seconds
Auditing took 40.9 seconds
Total runtime: 4118.37957 seconds

sweep_l3_cifar100_neggrad_plus — log-scale ROC



sweep_l3_cifar100_random_label

Level-3 U-RMIA (strong attack, train-then-unlearn references) · dataset cifar100

Configuration

algorithm : random_label
forget_size : 500
params : epochs=3, learning_rate=0.001

Accuracy

role	test_acc	forget_acc
original	0.2660	n/a
unlearned	0.1360	0.2560
retrained	0.2760	0.2760

AUC

role	conf_global	mentr_global	ulira	ulira_mentr	urmia_offline	urmia_online
original	0.7533	0.7576	0.5943	0.6147	0.7629	0.8006
unlearned	0.6472	0.6516	0.6109	0.6108	0.6493	0.6553
retrained	0.4892	0.4887	0.4657	0.4822	0.4900	0.4818

Two-sided AUC

role	conf_global	mentr_global	ulira	ulira_mentr	urmia_offline	urmia_online
original	0.7533	0.7576	0.5943	0.6147	0.7629	0.8006
unlearned	0.6472	0.6516	0.6109	0.6108	0.6493	0.6553
retrained	0.5108	0.5113	0.5343	0.5178	0.5100	0.5182

TPR @ 0.1% FPR

role	conf_global	mentr_global	ulira	ulira_mentr	urmia_offline	urmia_online
original	0.0000	0.0000	0.0040	0.0100	0.0160	0.0080
unlearned	0.0040	0.0040	0.0000	0.0000	0.0060	0.0000
retrained	0.0020	0.0020	0.0040	0.0000	0.0000	0.0120

TPR @ 1% FPR

role	conf_global	mentr_global	ulira	ulira_mentr	urmia_offline	urmia_online
original	0.0140	0.0140	0.0240	0.0380	0.0440	0.0580
unlearned	0.0040	0.0040	0.0020	0.0080	0.0220	0.0360
retrained	0.0080	0.0080	0.0100	0.0000	0.0140	0.0260

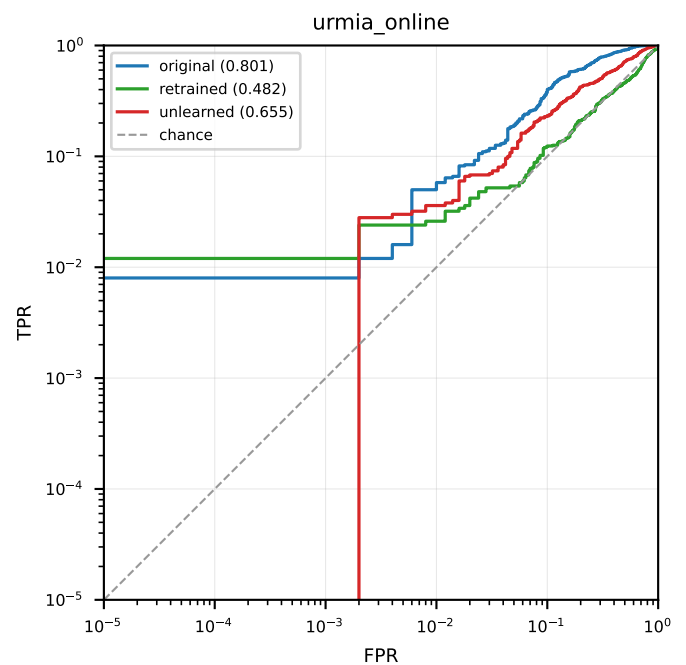
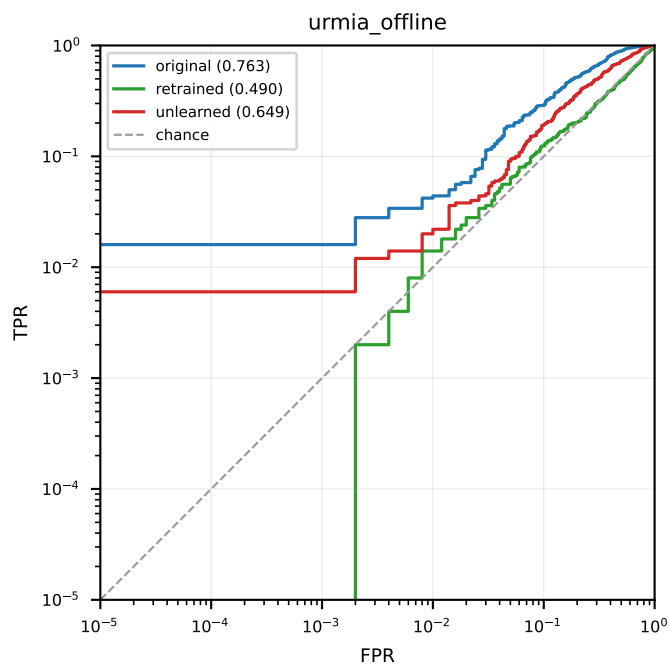
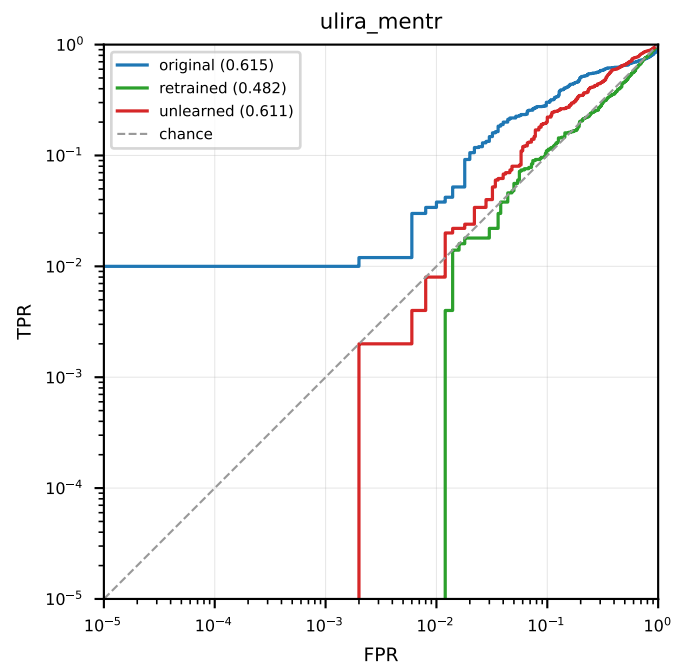
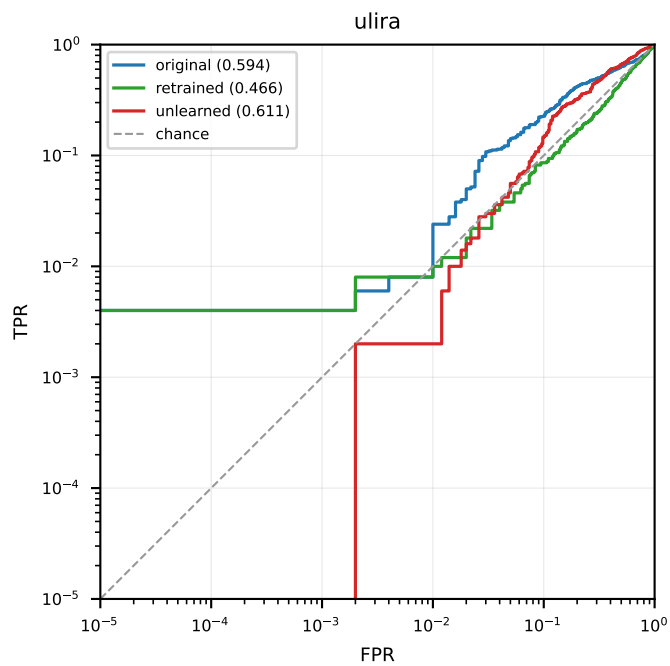
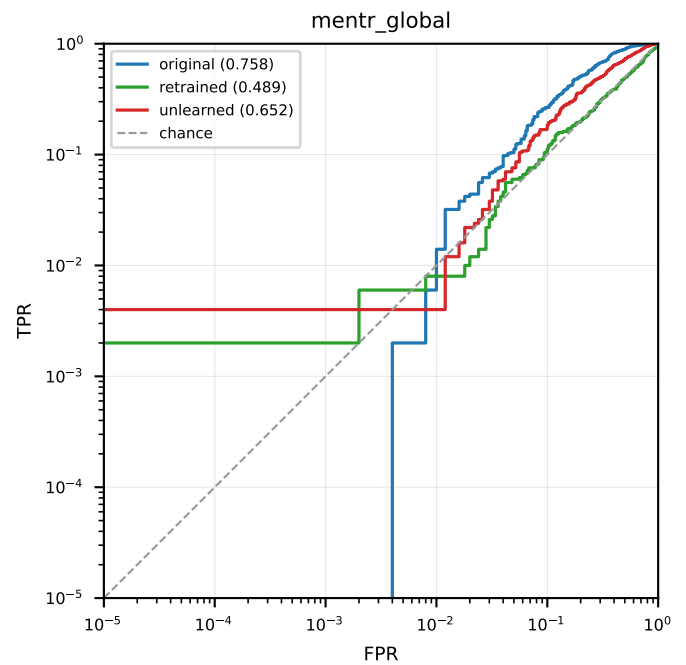
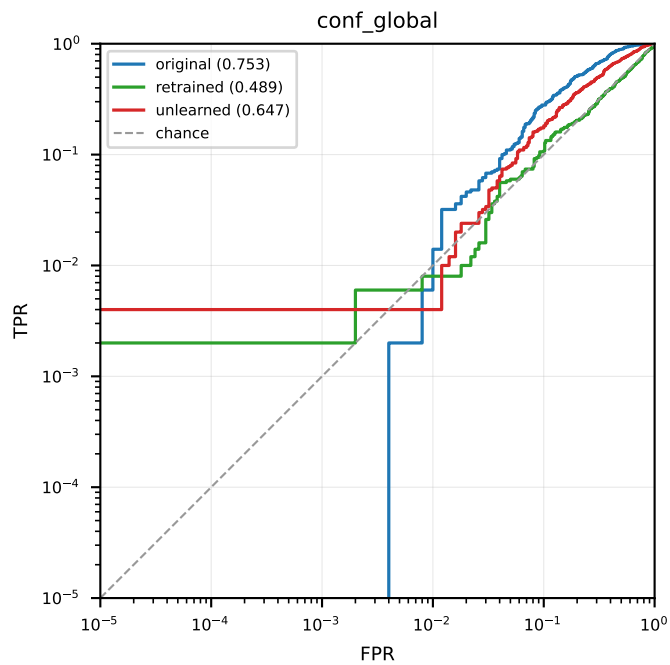
Positive control

dead_scorers : none
null_se : 0.01827
dead_threshold : 0.53653
elevated_threshold : 0.57307
original (two-sided) AUC: urmia_offline=0.7629, urmia_online=0.8006, ulira=0.5943, conf_global=0.7533, mentr_global=0.7576, ulira_mentr=0.6147

Timing

Loading dataset took 0.09471 seconds
Model loading/training took 3923.3 seconds
Preparing signals took 4.00322 seconds
Auditing took 41.4 seconds
Total runtime: 3968.92440 seconds

sweep_l3_cifar100_random_label — log-scale ROC



sweep_l3_cifar100_wig

Level-3 U-RMIA (strong attack, train-then-unlearn references) · dataset cifar100

Configuration

algorithm : wig
forget_size : 500
params : init_ratio=0.3, grad_batch_size=32, epochs=30, learning_rate=0.02

Accuracy

role	test_acc	forget_acc
original	0.2580	n/a
unlearned	0.2300	0.2680
retrained	0.3020	0.3140

AUC

role	conf_global	mentr_global	ulira	ulira_mentr	urmia_offline	urmia_online
original	0.7720	0.7743	0.5052	0.5038	0.7915	0.8326
unlearned	0.5191	0.5212	0.5127	0.5039	0.5197	0.5185
retrained	0.4864	0.4893	0.4922	0.4834	0.4846	0.4817

Two-sided AUC

role	conf_global	mentr_global	ulira	ulira_mentr	urmia_offline	urmia_online
original	0.7720	0.7743	0.5052	0.5038	0.7915	0.8326
unlearned	0.5191	0.5212	0.5127	0.5039	0.5197	0.5185
retrained	0.5136	0.5107	0.5078	0.5166	0.5154	0.5183

TPR @ 0.1% FPR

role	conf_global	mentr_global	ulira	ulira_mentr	urmia_offline	urmia_online
original	0.0220	0.0220	0.0120	0.0320	0.0080	0.0620
unlearned	0.0040	0.0040	0.0000	0.0000	0.0000	0.0000
retrained	0.0040	0.0040	0.0060	0.0040	0.0000	0.0020

TPR @ 1% FPR

role	conf_global	mentr_global	ulira	ulira_mentr	urmia_offline	urmia_online
original	0.0420	0.0400	0.0580	0.1080	0.1440	0.1440
unlearned	0.0180	0.0180	0.0000	0.0040	0.0360	0.0120
retrained	0.0160	0.0160	0.0080	0.0180	0.0440	0.0080

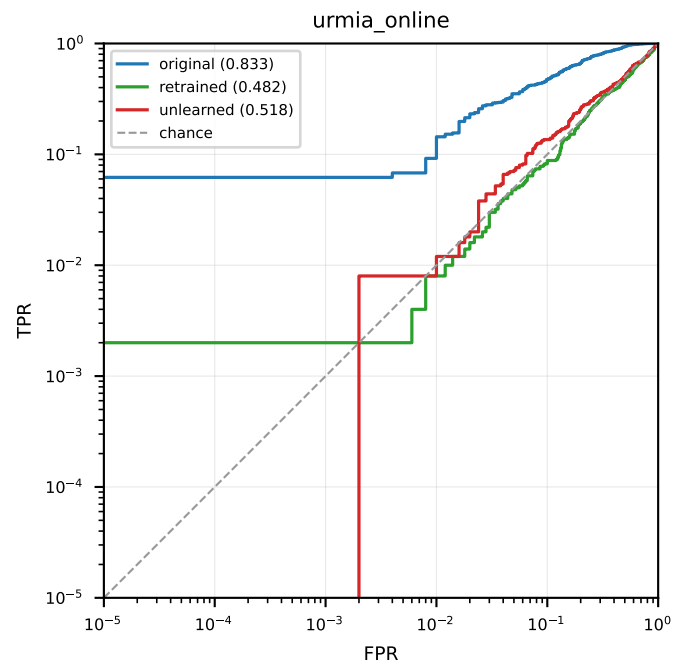
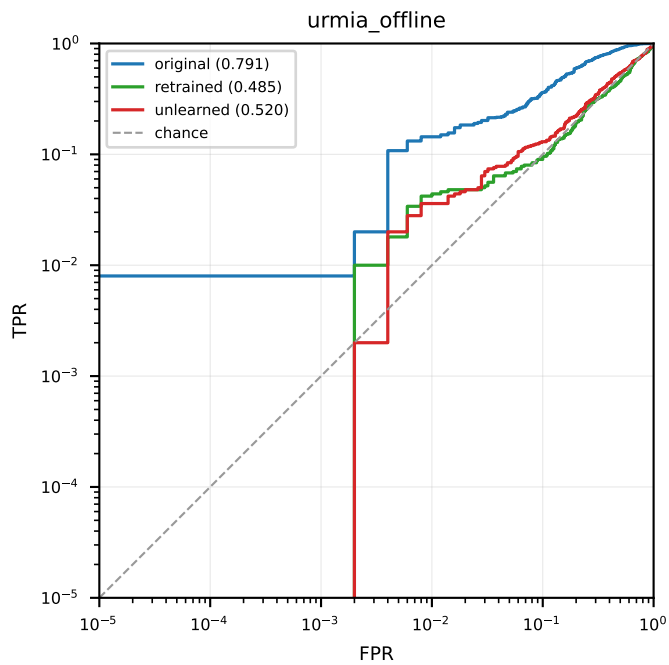
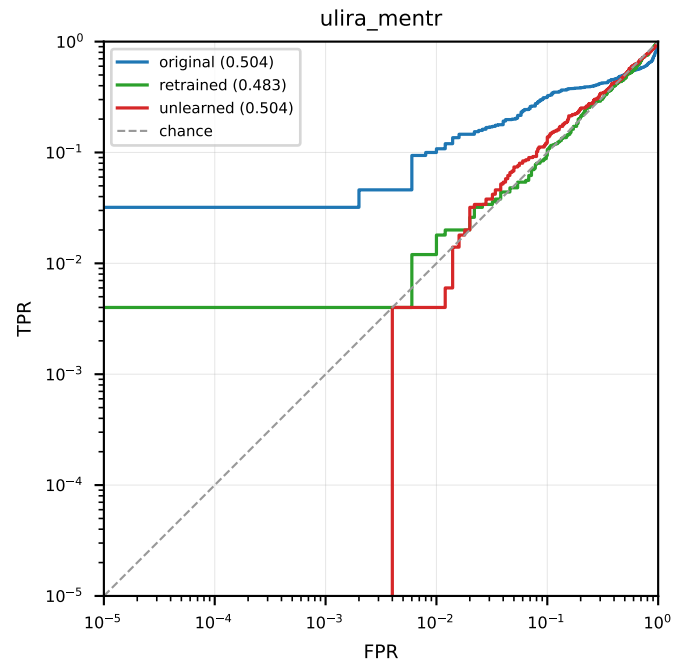
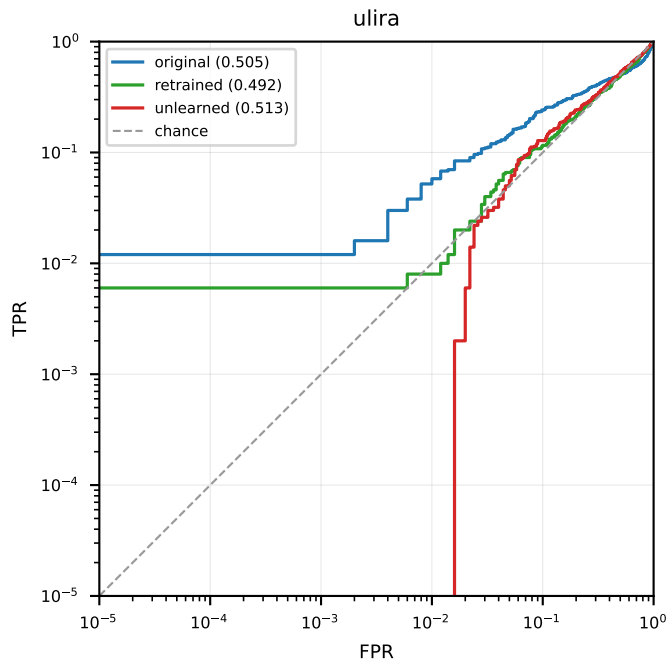
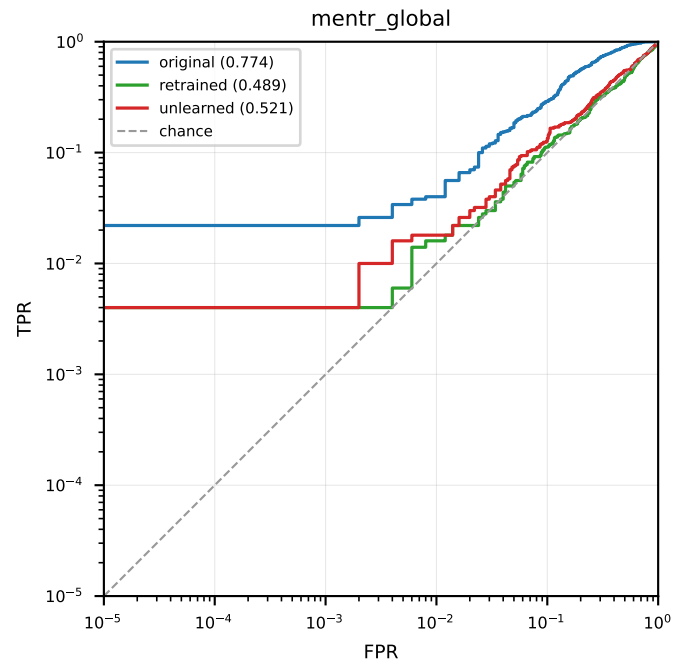
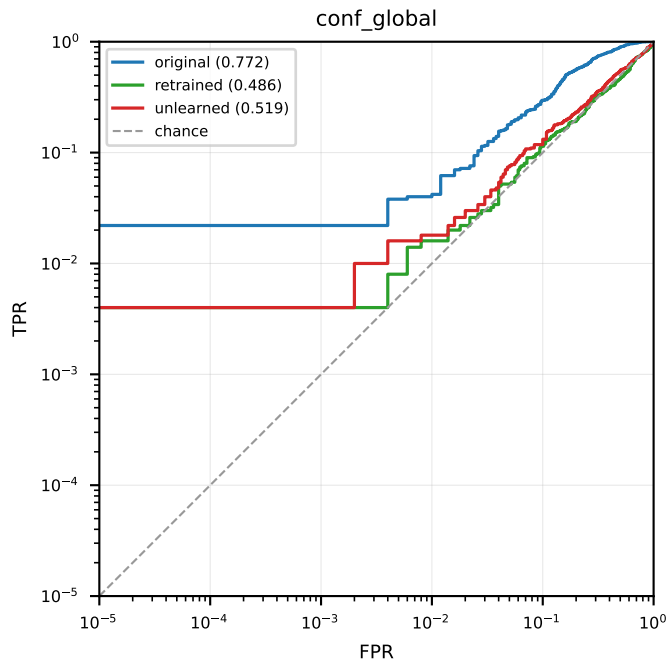
Positive control

dead_scorers : ['ulira', 'ulira_mentr']
null_se : 0.01827
dead_threshold : 0.53653
elevated_threshold : 0.57307
original (two-sided) AUC: urmia_offline=0.7915, urmia_online=0.8326, ulira=0.5052, conf_global=0.7720, mentr_global=0.7743, ulira_mentr=0.5038

Timing

Loading dataset took 0.09234 seconds
Model loading/training took 4680.2 seconds
Preparing signals took 4.20114 seconds
Auditing took 40.7 seconds
Total runtime: 4725.36278 seconds

sweep_l3_cifar100_wig — log-scale ROC



sweep_l3_cifar10_gaussian_noise

Level-3 U-RMIA (strong attack, train-then-unlearn references) · dataset cifar10

Configuration

algorithm : gaussian_noise
forget_size : 500
params : noise_std=0.003

Accuracy

role	test_acc	forget_acc
original	0.7220	n/a
unlearned	0.5920	0.7400
retrained	0.7110	0.7080

AUC

role	conf_global	mentr_global	ulira	ulira_mentr	urmia_offline	urmia_online
original	0.7080	0.7085	0.7137	0.7425	0.7641	0.7570
unlearned	0.5972	0.5924	0.6149	0.6119	0.6377	0.6309
retrained	0.4870	0.4874	0.5011	0.5114	0.5107	0.5127

Two-sided AUC

role	conf_global	mentr_global	ulira	ulira_mentr	urmia_offline	urmia_online
original	0.7080	0.7085	0.7137	0.7425	0.7641	0.7570
unlearned	0.5972	0.5924	0.6149	0.6119	0.6377	0.6309
retrained	0.5130	0.5126	0.5011	0.5114	0.5107	0.5127

TPR @ 0.1% FPR

role	conf_global	mentr_global	ulira	ulira_mentr	urmia_offline	urmia_online
original	0.0000	0.0000	0.0020	0.0300	0.0120	0.0000
unlearned	0.0000	0.0000	0.0000	0.0040	0.0040	0.0000
retrained	0.0000	0.0000	0.0040	0.0000	0.0000	0.0000

TPR @ 1% FPR

role	conf_global	mentr_global	ulira	ulira_mentr	urmia_offline	urmia_online
original	0.0080	0.0060	0.0580	0.0540	0.0520	0.0280
unlearned	0.0160	0.0160	0.0320	0.0540	0.0360	0.0340
retrained	0.0100	0.0100	0.0040	0.0060	0.0080	0.0040

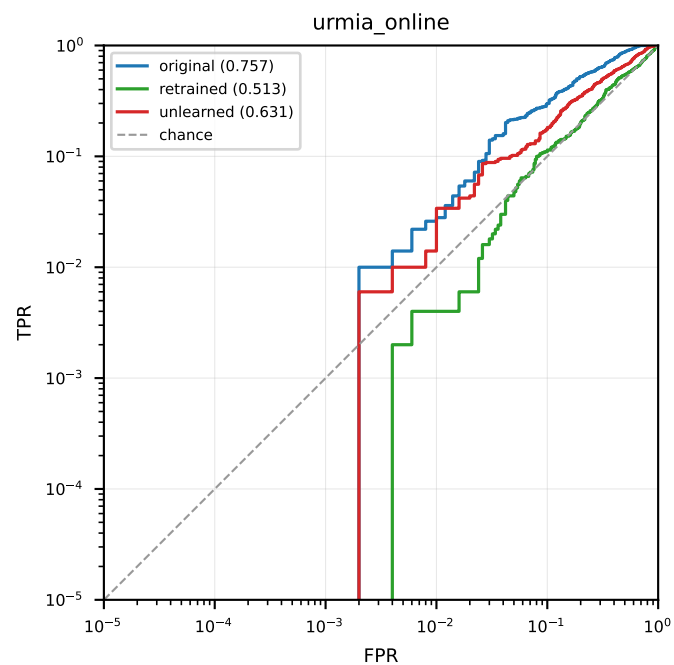
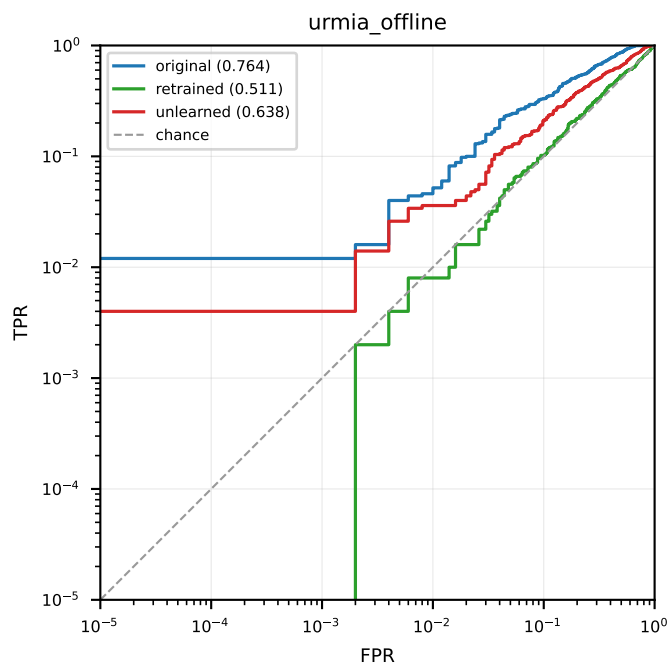
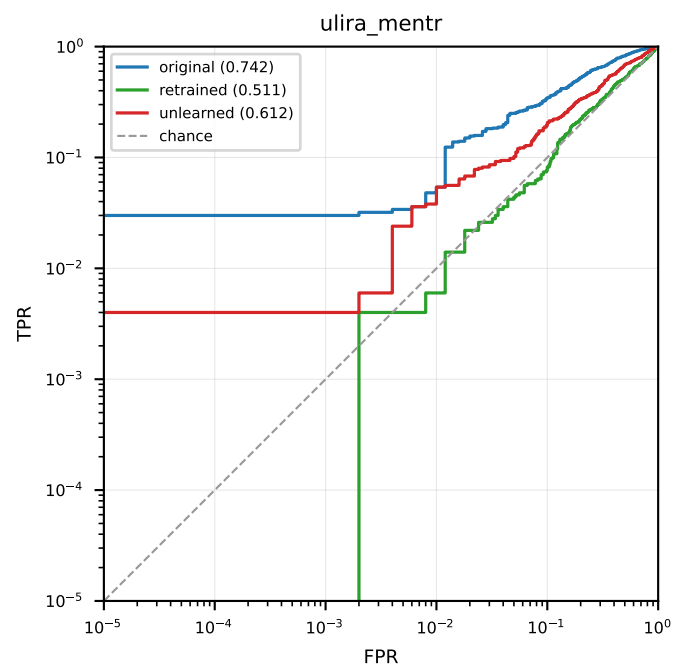
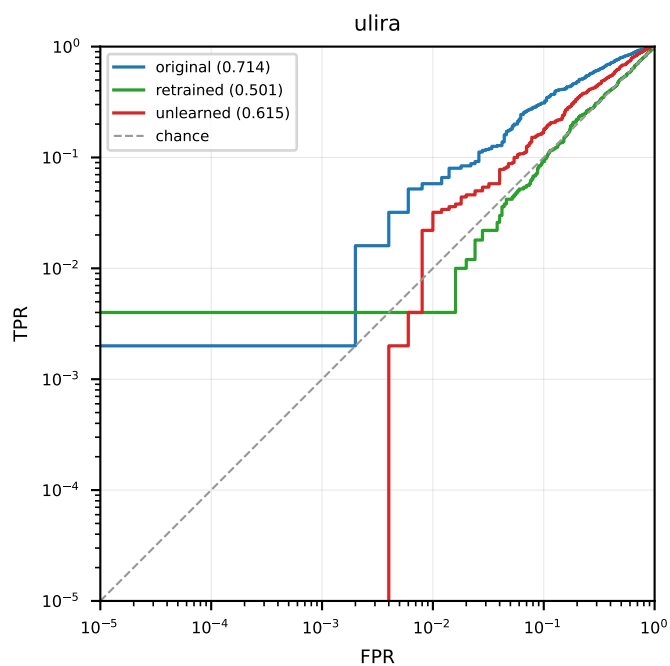
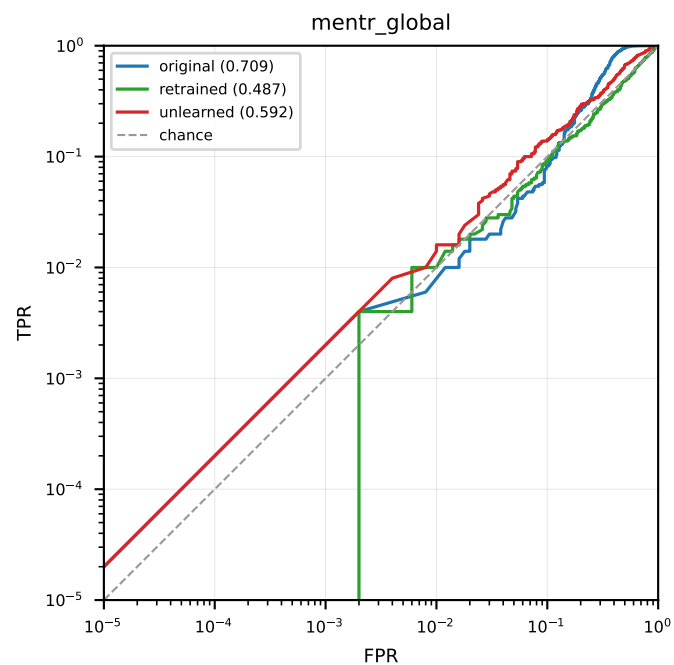
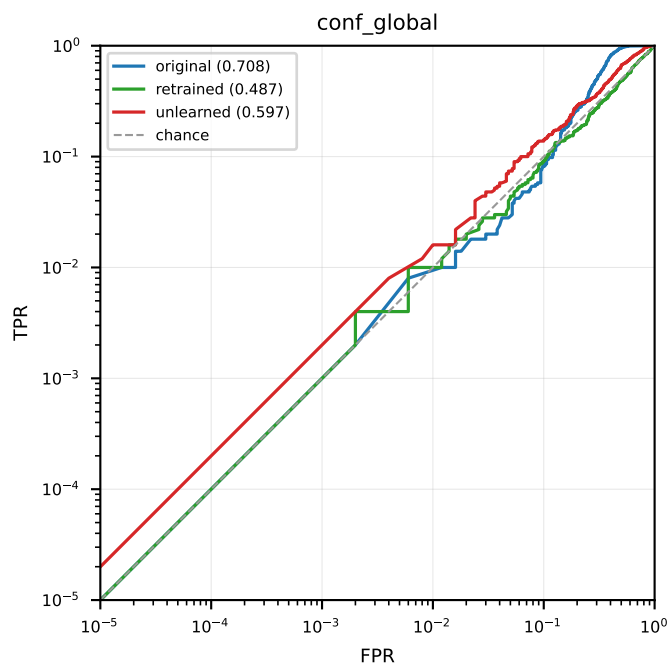
Positive control

dead_scorers : none
null_se : 0.01827
dead_threshold : 0.53653
elevated_threshold : 0.57307
original (two-sided) AUC: urmia_offline=0.7641, urmia_online=0.7570, ulira=0.7137, conf_global=0.7080, mentr_global=0.7085, ulira_mentr=0.7425

Timing

Loading dataset took 0.14770 seconds
Model loading/training took 3936.5 seconds
Preparing signals took 7.79120 seconds
Auditing took 58.2 seconds
Total runtime: 4002.79097 seconds

sweep_l3_cifar10_gaussian_noise — log-scale ROC



sweep_l3_cifar10_neggrad

Level-3 U-RMIA (strong attack, train-then-unlearn references) · dataset cifar10

Configuration

algorithm : neggrad
forget_size : 500
params : epochs=3, learning_rate=0.001

Accuracy

role	test_acc	forget_acc
original	0.7140	n/a
unlearned	0.7040	1.0000
retrained	0.7070	0.6940

AUC

role	conf_global	mentr_global	ulira	ulira_mentr	urmia_offline	urmia_online
original	0.7015	0.7017	0.7499	0.7874	0.7783	0.7901
unlearned	0.7021	0.7026	0.7624	0.7969	0.7751	0.7880
retrained	0.4878	0.4880	0.5168	0.5161	0.4934	0.4955

Two-sided AUC

role	conf_global	mentr_global	ulira	ulira_mentr	urmia_offline	urmia_online
original	0.7015	0.7017	0.7499	0.7874	0.7783	0.7901
unlearned	0.7021	0.7026	0.7624	0.7969	0.7751	0.7880
retrained	0.5122	0.5120	0.5168	0.5161	0.5066	0.5045

TPR @ 0.1% FPR

role	conf_global	mentr_global	ulira	ulira_mentr	urmia_offline	urmia_online
original	0.0000	0.0000	0.0040	0.0060	0.0260	0.0340
unlearned	0.0000	0.0000	0.0100	0.0100	0.0260	0.0340
retrained	0.0040	0.0040	0.0040	0.0020	0.0000	0.0040

TPR @ 1% FPR

role	conf_global	mentr_global	ulira	ulira_mentr	urmia_offline	urmia_online
original	0.0100	0.0060	0.0480	0.0860	0.0800	0.1040
unlearned	0.0160	0.0160	0.0440	0.0880	0.0720	0.0980
retrained	0.0060	0.0060	0.0280	0.0200	0.0100	0.0200

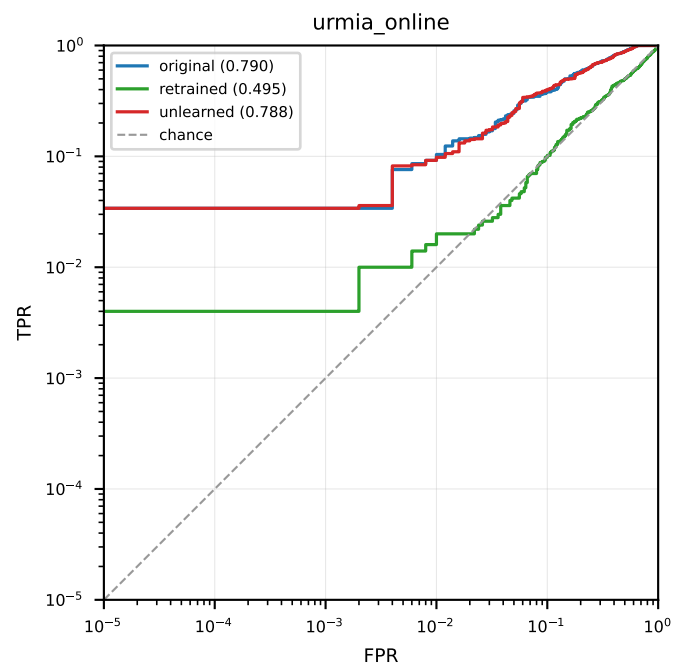
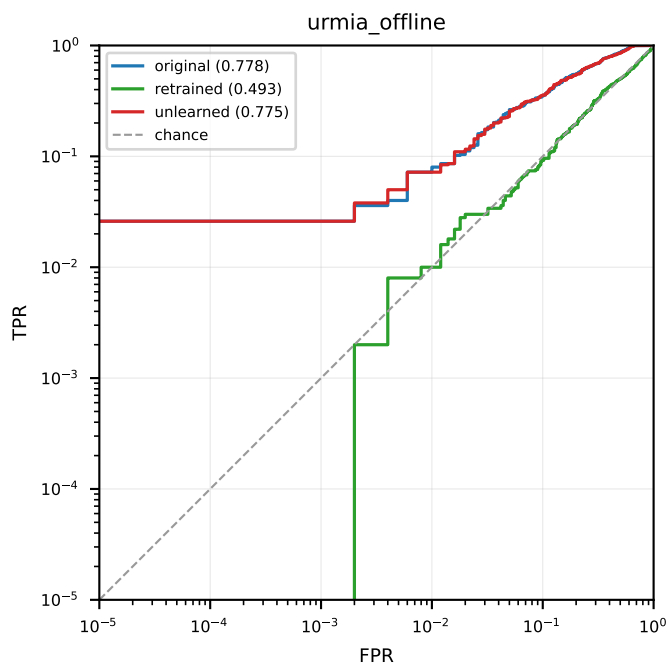
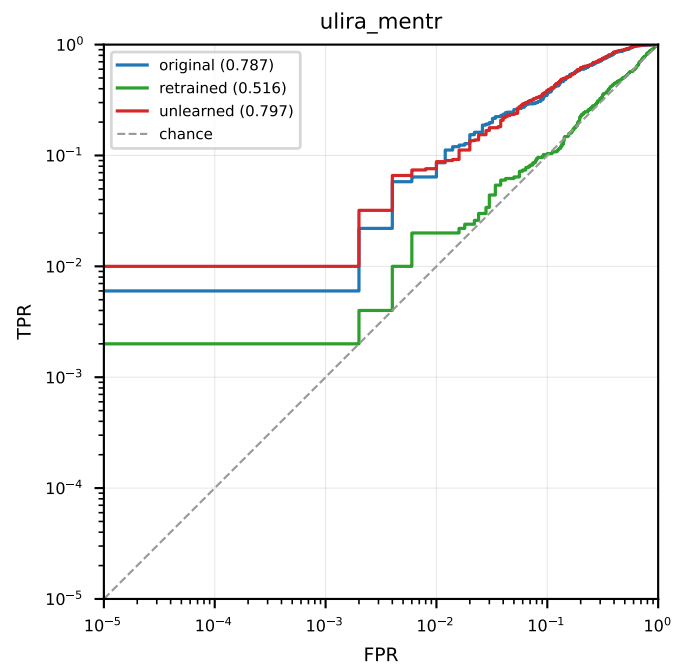
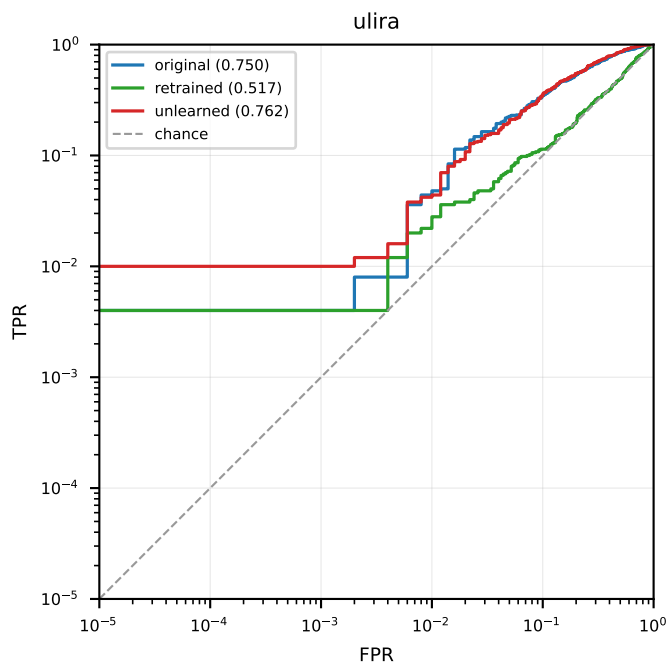
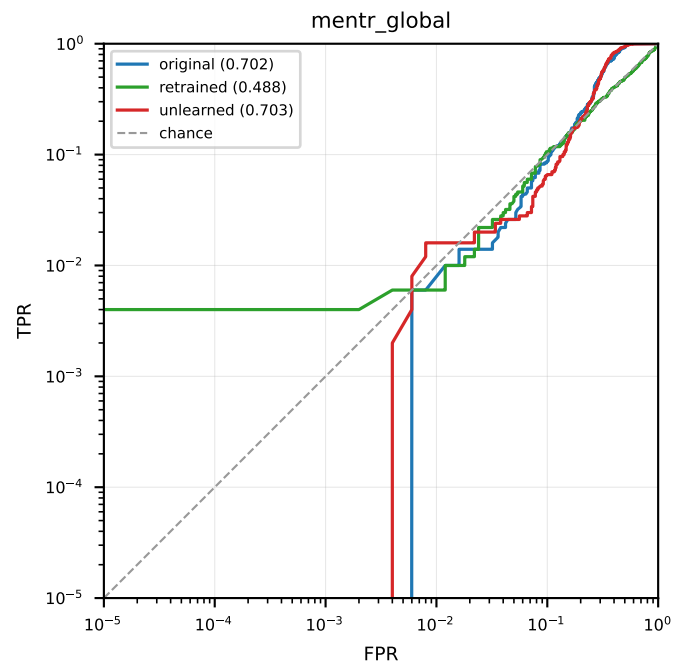
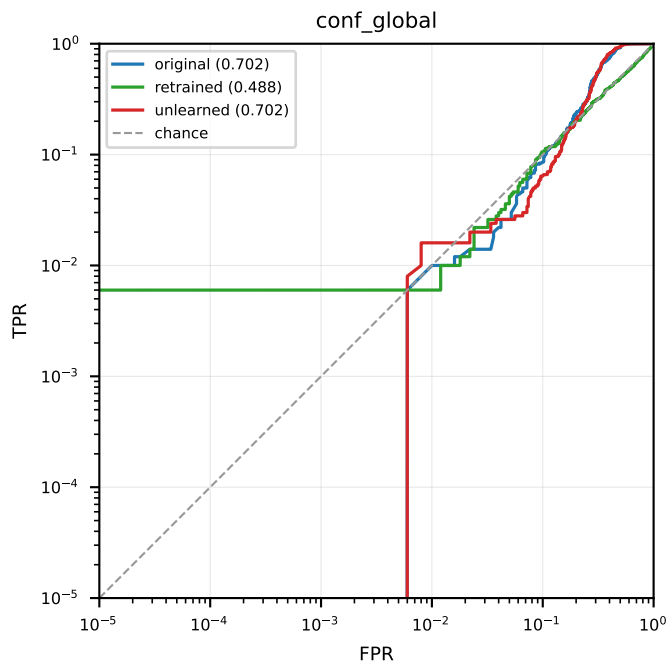
Positive control

dead_scorers : none
null_se : 0.01827
dead_threshold : 0.53653
elevated_threshold : 0.57307
original (two-sided) AUC: urmia_offline=0.7783, urmia_online=0.7901, ulira=0.7499, conf_global=0.7015, mentr_global=0.7017, ulira_mentr=0.7874

Timing

Loading dataset took 0.25922 seconds
Model loading/training took 4042.5 seconds
Preparing signals took 5.27856 seconds
Auditing took 47.1 seconds
Total runtime: 4095.61427 seconds

sweep_I3_cifar10_neggrad — log-scale ROC



sweep_l3_cifar10_neggrad_plus

Level-3 U-RMIA (strong attack, train-then-unlearn references) · dataset cifar10

Configuration

algorithm : neggrad_plus
forget_size : 500
params : epochs=2, learning_rate=0.002, forget_coef=0.5, retain_coef=1.0, forget_loss_cap=1.16

Accuracy

role	test_acc	forget_acc
original	0.7060	n/a
unlearned	0.7120	0.8200
retrained	0.7070	0.7100

AUC

role	conf_global	mentr_global	ulira	ulira_mentr	urmia_offline	urmia_online
original	0.6849	0.6853	0.7110	0.7592	0.7825	0.7754
unlearned	0.5464	0.5455	0.6485	0.6558	0.6163	0.6150
retrained	0.4897	0.4902	0.5038	0.5269	0.5096	0.5111

Two-sided AUC

role	conf_global	mentr_global	ulira	ulira_mentr	urmia_offline	urmia_online
original	0.6849	0.6853	0.7110	0.7592	0.7825	0.7754
unlearned	0.5464	0.5455	0.6485	0.6558	0.6163	0.6150
retrained	0.5103	0.5098	0.5038	0.5269	0.5096	0.5111

TPR @ 0.1% FPR

role	conf_global	mentr_global	ulira	ulira_mentr	urmia_offline	urmia_online
original	0.0000	0.0000	0.0200	0.0000	0.0300	0.0120
unlearned	0.0000	0.0000	0.0140	0.0000	0.0120	0.0020
retrained	0.0000	0.0000	0.0000	0.0040	0.0020	0.0020

TPR @ 1% FPR

role	conf_global	mentr_global	ulira	ulira_mentr	urmia_offline	urmia_online
original	0.0040	0.0040	0.0900	0.0980	0.0980	0.1220
unlearned	0.0060	0.0060	0.0640	0.0620	0.0540	0.0600
retrained	0.0040	0.0040	0.0100	0.0080	0.0080	0.0120

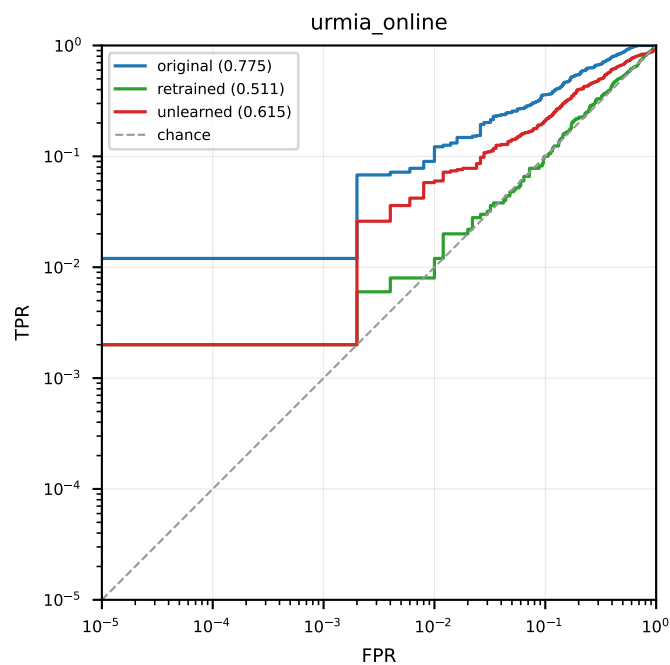
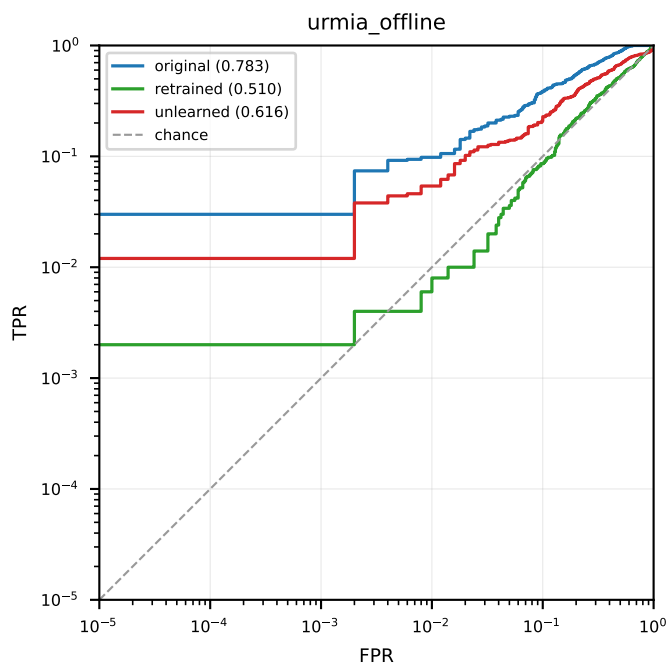
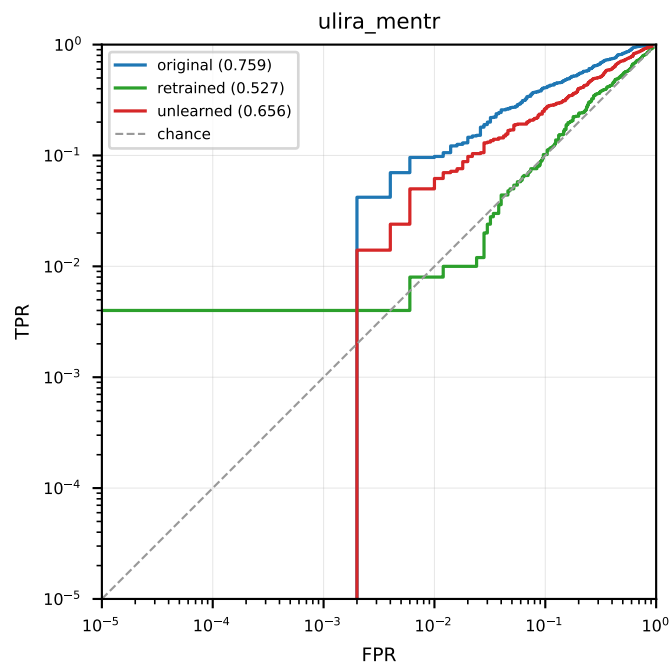
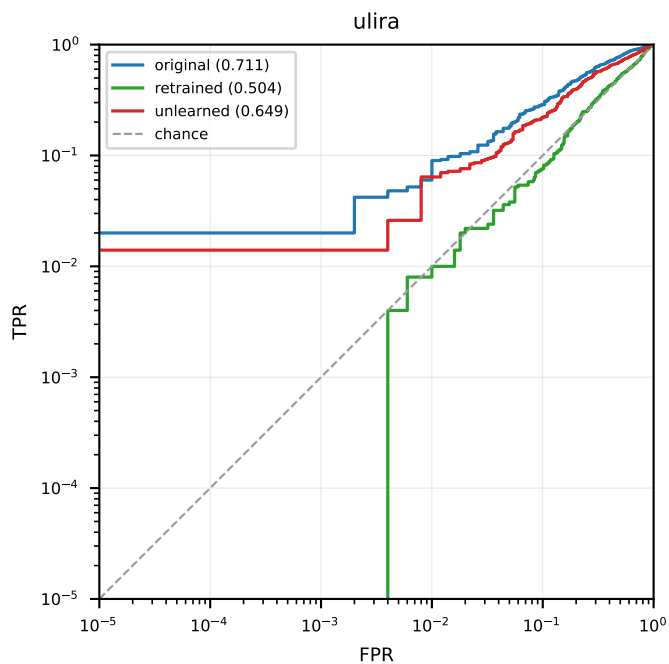
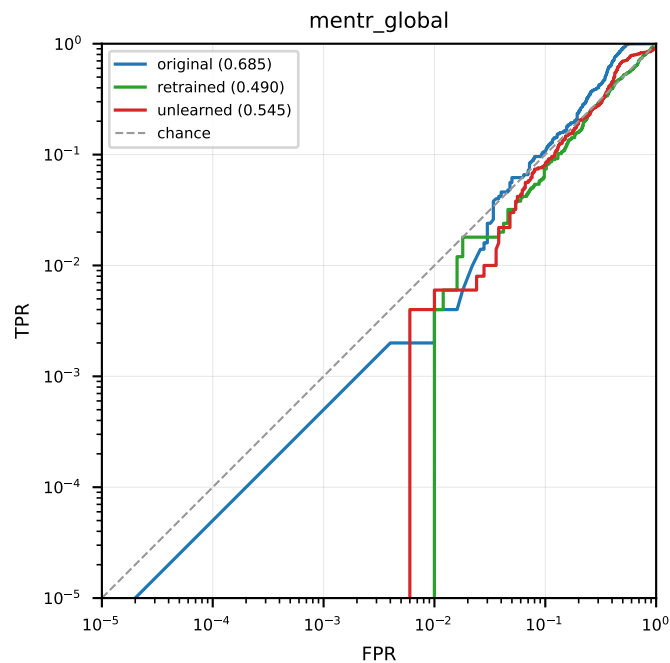
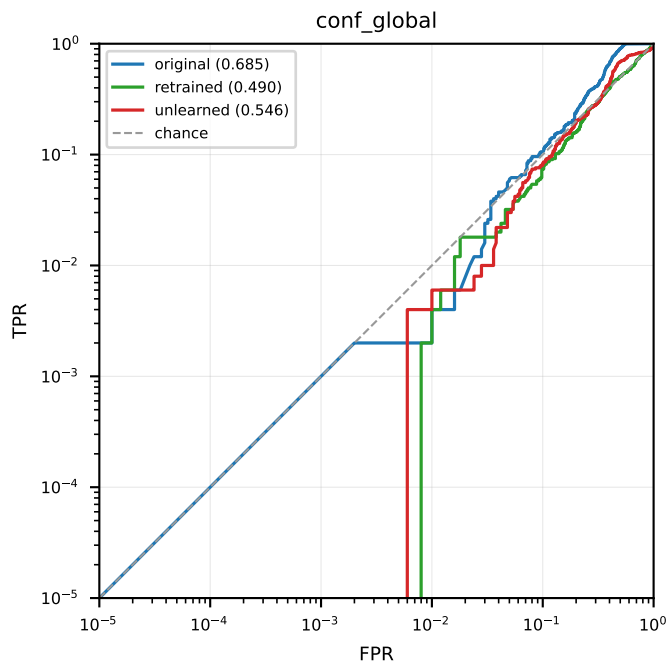
Positive control

dead_scorers : none
null_se : 0.01827
dead_threshold : 0.53653
elevated_threshold : 0.57307
original (two-sided) AUC: urmia_offline=0.7825, urmia_online=0.7754, ulira=0.7110, conf_global=0.6849, zentr_global=0.6853, ulira_mentr=0.7592

Timing

Loading dataset took 0.09639 seconds
Model loading/training took 4042.0 seconds
Preparing signals took 6.14199 seconds
Auditing took 42.7 seconds
Total runtime: 4091.00158 seconds

sweep_l3_cifar10_neggrad_plus — log-scale ROC



sweep_l3_cifar10_random_label

Level-3 U-RMIA (strong attack, train-then-unlearn references) · dataset cifar10

Configuration

algorithm : random_label
forget_size : 500
params : epochs=3, learning_rate=0.001

Accuracy

role	test_acc	forget_acc

original	0.7140	n/a
unlearned	0.2880	0.2440
retrained	0.7100	0.7060

AUC

role	conf_global	mentr_global	ulira	ulira_mentr	urmia_offline	urmia_online

original	0.6881	0.6881	0.5312	0.5268	0.6978	0.6265
unlearned	0.5039	0.5062	0.5635	0.5386	0.5070	0.5109
retrained	0.4865	0.4868	0.4748	0.4785	0.4750	0.4835

Two-sided AUC

role	conf_global	mentr_global	ulira	ulira_mentr	urmia_offline	urmia_online

original	0.6881	0.6881	0.5312	0.5268	0.6978	0.6265
unlearned	0.5039	0.5062	0.5635	0.5386	0.5070	0.5109
retrained	0.5135	0.5132	0.5252	0.5215	0.5250	0.5165

TPR @ 0.1% FPR

role	conf_global	mentr_global	ulira	ulira_mentr	urmia_offline	urmia_online

original	0.0000	0.0000	0.0040	0.0040	0.0120	0.0020
unlearned	0.0000	0.0000	0.0000	0.0000	0.0040	0.0000
retrained	0.0040	0.0020	0.0040	0.0040	0.0060	0.0020

TPR @ 1% FPR

role	conf_global	mentr_global	ulira	ulira_mentr	urmia_offline	urmia_online

original	0.0040	0.0040	0.0460	0.0440	0.0300	0.0080
unlearned	0.0040	0.0040	0.0040	0.0020	0.0040	0.0040
retrained	0.0080	0.0080	0.0140	0.0160	0.0180	0.0060

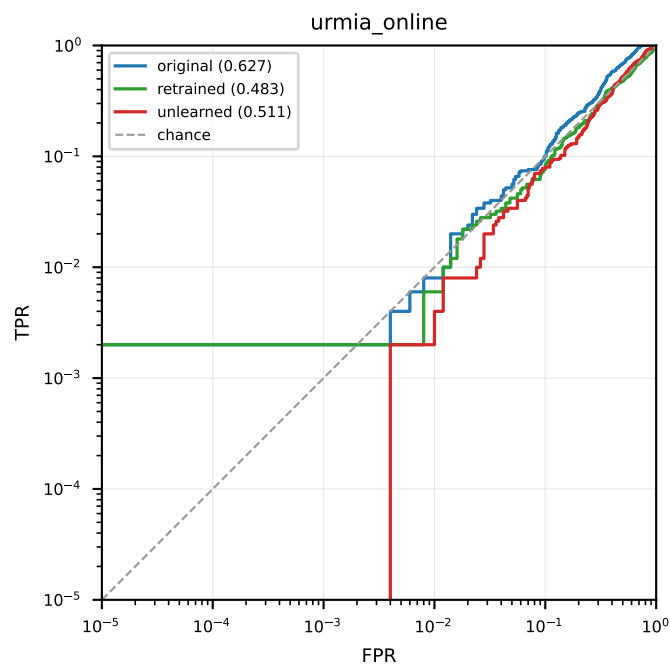
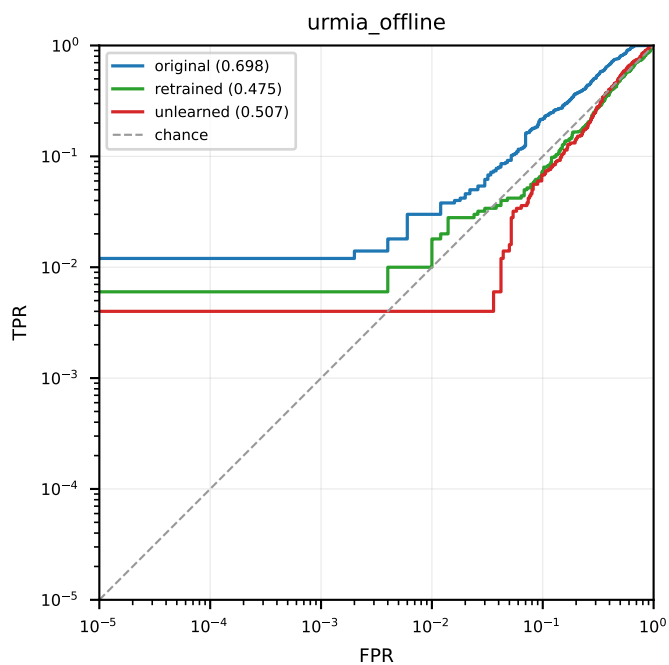
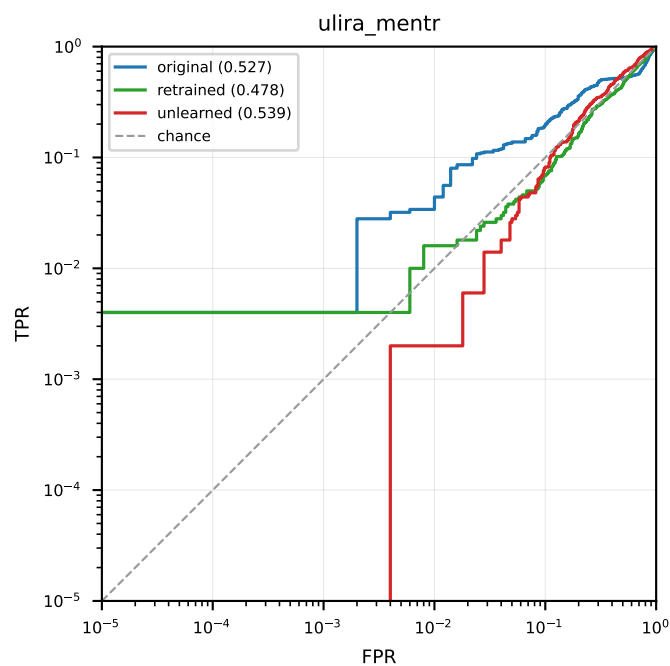
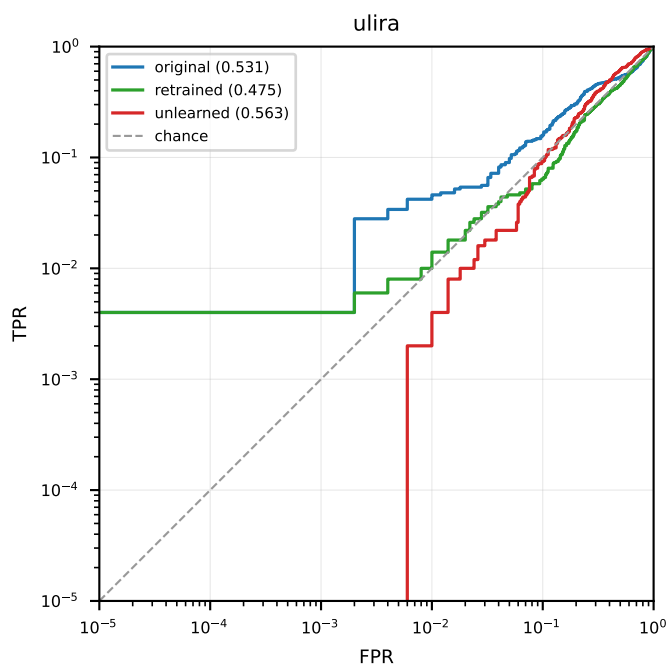
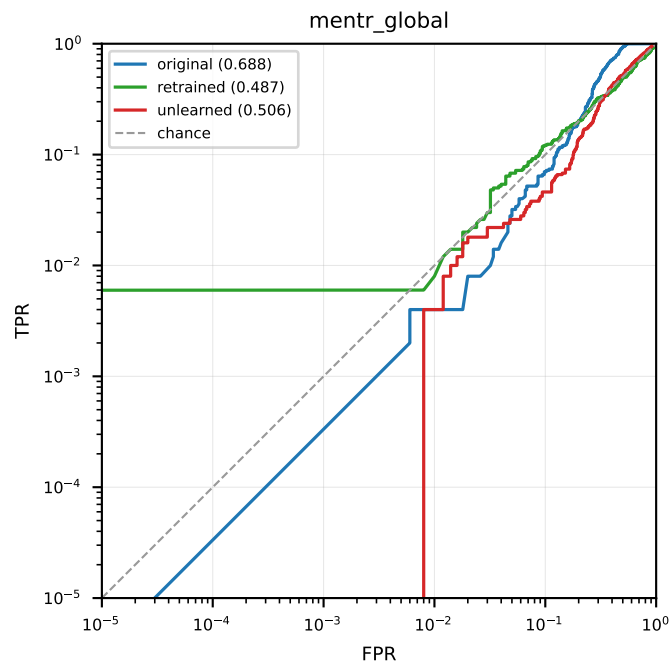
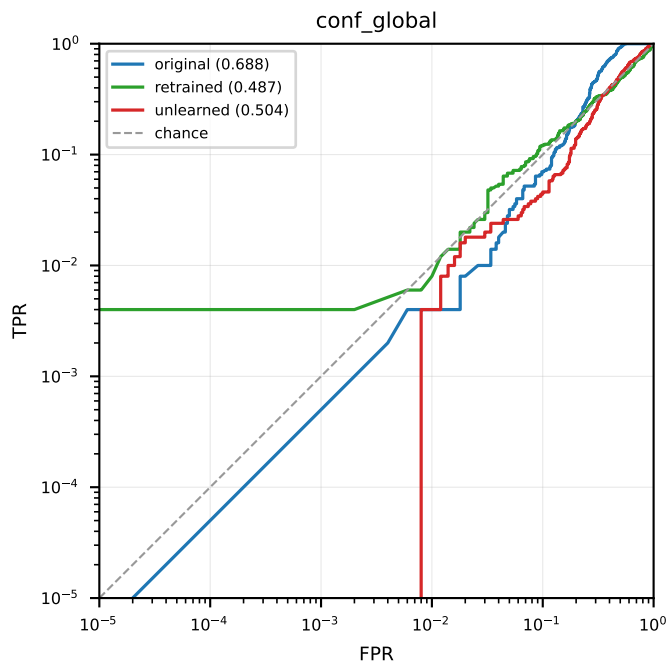
Positive control

dead_scorers : ['ulira', 'ulira_mentr']
null_se : 0.01827
dead_threshold : 0.53653
elevated_threshold : 0.57307
original (two-sided) AUC: urmia_offline=0.6978, urmia_online=0.6265, ulira=0.5312, conf_global=0.6881, mentr_global=0.6881, ulira_mentr=0.5268

Timing

Loading dataset took 0.18579 seconds
Model loading/training took 3950.2 seconds
Preparing signals took 4.12284 seconds
Auditing took 40.9 seconds
Total runtime: 3995.59533 seconds

sweep_l3_cifar10_random_label — log-scale ROC



sweep_l3_cifar10_wig

Level-3 U-RMIA (strong attack, train-then-unlearn references) · dataset cifar10

Configuration

algorithm : wig
forget_size : 500
params : init_ratio=0.3, grad_batch_size=32, epochs=30, learning_rate=0.02

Accuracy

role	test_acc	forget_acc
original	0.7220	n/a
unlearned	0.6380	0.6140
retrained	0.7070	0.7160

AUC

role	conf_global	mentr_global	ulira	ulira_mentr	urmia_offline	urmia_online
original	0.6908	0.6919	0.5318	0.5296	0.7683	0.7316
unlearned	0.4792	0.4781	0.5202	0.5104	0.4979	0.5046
retrained	0.5037	0.5049	0.5319	0.5348	0.5285	0.5324

Two-sided AUC

role	conf_global	mentr_global	ulira	ulira_mentr	urmia_offline	urmia_online
original	0.6908	0.6919	0.5318	0.5296	0.7683	0.7316
unlearned	0.5208	0.5219	0.5202	0.5104	0.5021	0.5046
retrained	0.5037	0.5049	0.5319	0.5348	0.5285	0.5324

TPR @ 0.1% FPR

role	conf_global	mentr_global	ulira	ulira_mentr	urmia_offline	urmia_online
original	0.0000	0.0000	0.0000	0.0000	0.0020	0.0200
unlearned	0.0000	0.0000	0.0040	0.0000	0.0000	0.0000
retrained	0.0040	0.0060	0.0020	0.0020	0.0020	0.0000

TPR @ 1% FPR

role	conf_global	mentr_global	ulira	ulira_mentr	urmia_offline	urmia_online
original	0.0020	0.0020	0.0260	0.0640	0.0680	0.0640
unlearned	0.0080	0.0140	0.0120	0.0220	0.0100	0.0120
retrained	0.0120	0.0120	0.0140	0.0160	0.0060	0.0060

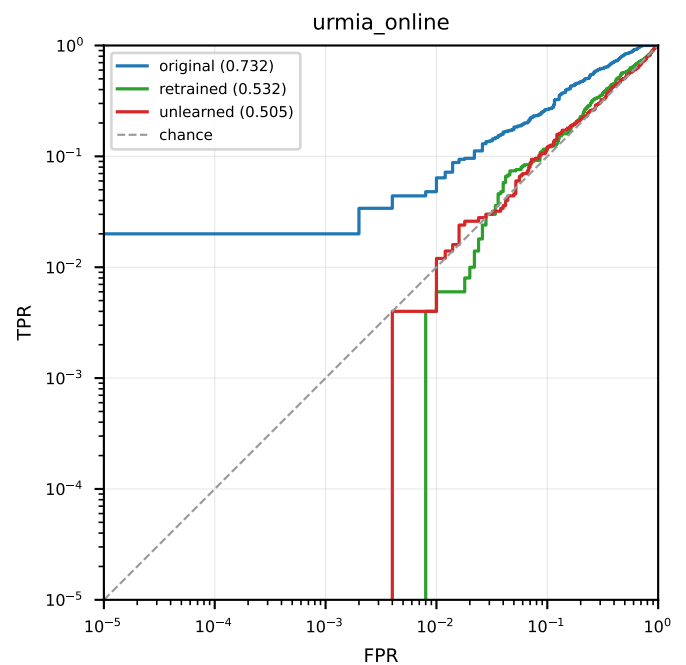
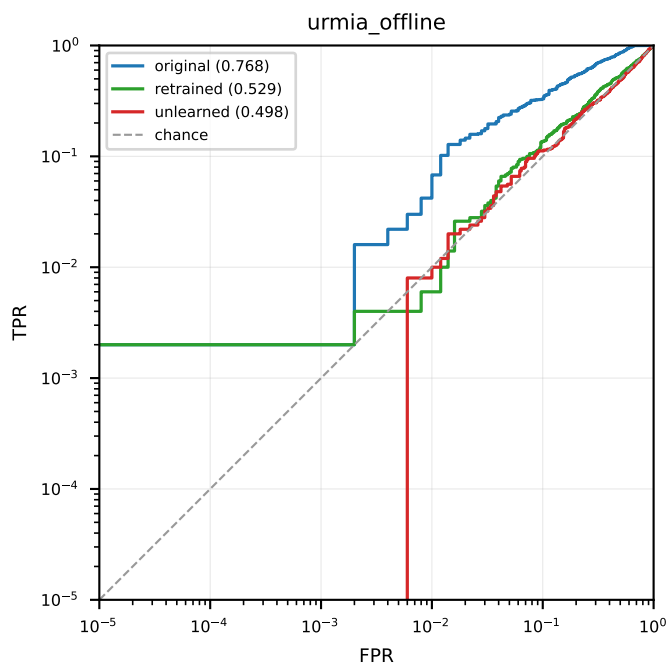
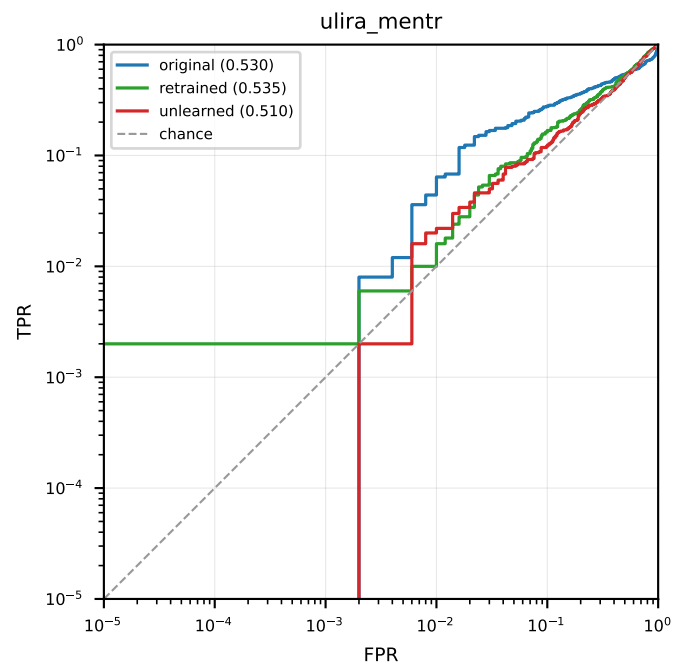
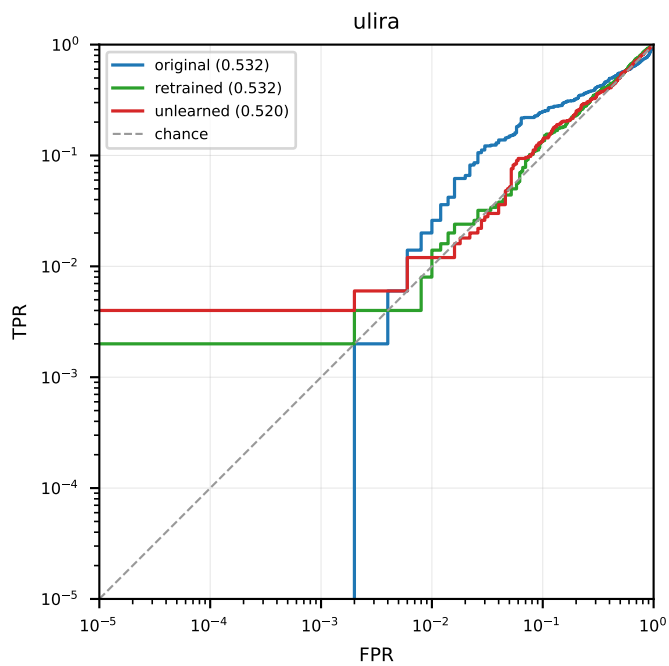
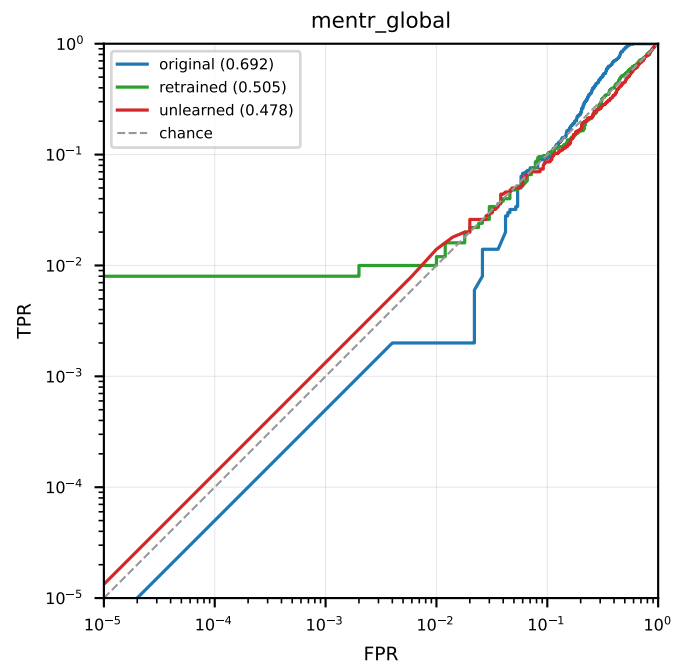
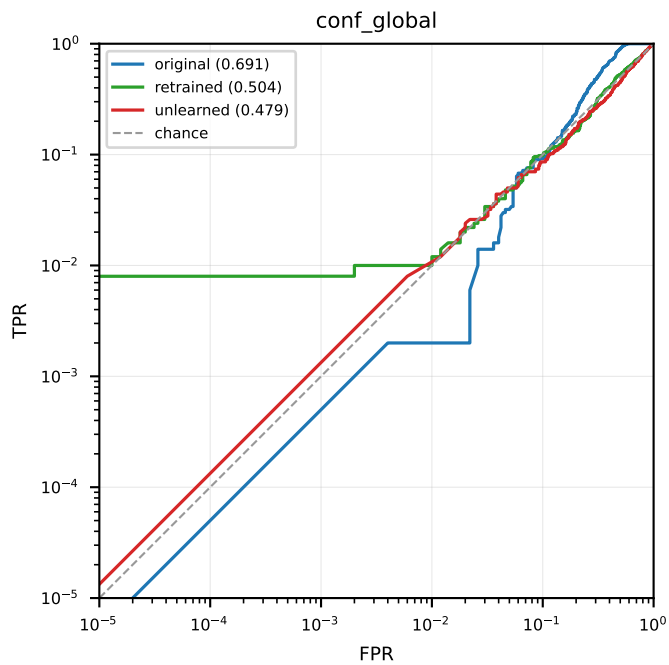
Positive control

dead_scorers : ['ulira', 'ulira_mentr']
null_se : 0.01827
dead_threshold : 0.53653
elevated_threshold : 0.57307
original (two-sided) AUC: urmia_offline=0.7683, urmia_online=0.7316, ulira=0.5318, conf_global=0.6908, mentr_global=0.6919, ulira_mentr=0.5296

Timing

Loading dataset took 0.10847 seconds
Model loading/training took 4677.2 seconds
Preparing signals took 6.52554 seconds
Auditing took 47.1 seconds
Total runtime: 4731.09189 seconds

sweep_l3_cifar10_wig — log-scale ROC



OLA (Outer Layer Attenuation)

2 run(s)

Comparison across runs

Two-sided AUC on the OUTER subset (the attenuated points), one row per role.

run / role	conf_global	lira	rmia_offline	rmia_online
cifar10/vanilla	0.7772	0.9941	0.9952	0.9961
cifar10/ola	0.8175	0.7991	0.7463	0.7623
cifar10/onion	0.8238	0.8140	0.7682	0.7828
cifar100/vanilla	0.8865	0.9878	0.9484	0.9678
cifar100/ola	0.7148	0.6597	0.6998	0.7117
cifar100/onion	0.7720	0.7309	0.7630	0.7712

sweep_ola_cifar10

OLA (Outer Layer Attenuation) · dataset cifar10

Configuration

outer_size : 1000
p_gen : 0.5
p_mem : 0.2
selection_fpr: 0.01
mask_seed : 12345
test_size : 5000

Accuracy

role	inner_acc	outer_acc	p_gen	p_mem	test_acc	train_size
vanilla	1.0000	1.0000	n/a	n/a	0.6336	25000
ola	1.0000	0.1610	0.5000	0.2000	0.6720	25000
onion	1.0000	0.1240	n/a	n/a	0.6334	24000

Subset: outer

AUC				
role	conf_global	lira	rmia_offline	rmia_online
vanilla	0.7772	0.9941	0.9952	0.9961
ola	0.1825	0.2009	0.2537	0.2377
onion	0.1762	0.1860	0.2318	0.2172

Two-sided AUC				
role	conf_global	lira	rmia_offline	rmia_online
vanilla	0.7772	0.9941	0.9952	0.9961
ola	0.8175	0.7991	0.7463	0.7623
onion	0.8238	0.8140	0.7682	0.7828

TPR @ 0.1% FPR				
role	conf_global	lira	rmia_offline	rmia_online
vanilla	0.0000	0.5300	0.5310	0.5770
ola	0.0000	0.0040	0.0090	0.0070
onion	0.0000	0.0000	0.0010	0.0010

Subset: inner

AUC				
role	conf_global	lira	rmia_offline	rmia_online
vanilla	0.8154	0.9078	0.8292	0.8439
ola	0.7814	0.8526	0.7927	0.8047
onion	0.8296	0.9213	0.8355	0.8503

Two-sided AUC				
role	conf_global	lira	rmia_offline	rmia_online
vanilla	0.8154	0.9078	0.8292	0.8439
ola	0.7814	0.8526	0.7927	0.8047
onion	0.8296	0.9213	0.8355	0.8503

TPR @ 0.1% FPR				
role	conf_global	lira	rmia_offline	rmia_online
vanilla	0.0000	0.0698	0.0560	0.0634
ola	0.0002	0.0437	0.0399	0.0419
onion	0.0000	0.1073	0.0607	0.0698

Subset: all_members

AUC				
role	conf_global	lira	rmia_offline	rmia_online
vanilla	0.8139	0.9112	0.8359	0.8500
ola	0.7574	0.8265	0.7712	0.7820
onion	0.8035	0.8919	0.8114	0.8250

Two-sided AUC

role	conf_global	lira	rmia_offline	rmia_online
vanilla	0.8139	0.9112	0.8359	0.8500
ola	0.7574	0.8265	0.7712	0.7820
onion	0.8035	0.8919	0.8114	0.8250

TPR @ 0.1% FPR

role	conf_global	lira	rmia_offline	rmia_online
vanilla	0.0000	0.0882	0.0750	0.0839
ola	0.0002	0.0421	0.0386	0.0405
onion	0.0000	0.1030	0.0583	0.0671

Excess over the onion floor (two-sided AUC difference)

outer (null SE 0.00931)

	conf_global	lira	rmia_offline	rmia_online
excess	-0.0063	-0.0149	-0.0219	-0.0205
in null SE	-0.6718	-1.6010	-2.3489	-2.1997

inner (null SE 0.00261)

	conf_global	lira	rmia_offline	rmia_online
excess	-0.0482	-0.0687	-0.0428	-0.0456
in null SE	-18.4858	-26.3409	-16.4099	-17.4771

all_members (null SE 0.00258)

	conf_global	lira	rmia_offline	rmia_online
excess	-0.0460	-0.0654	-0.0402	-0.0430
in null SE	-17.8334	-25.3185	-15.5780	-16.6346

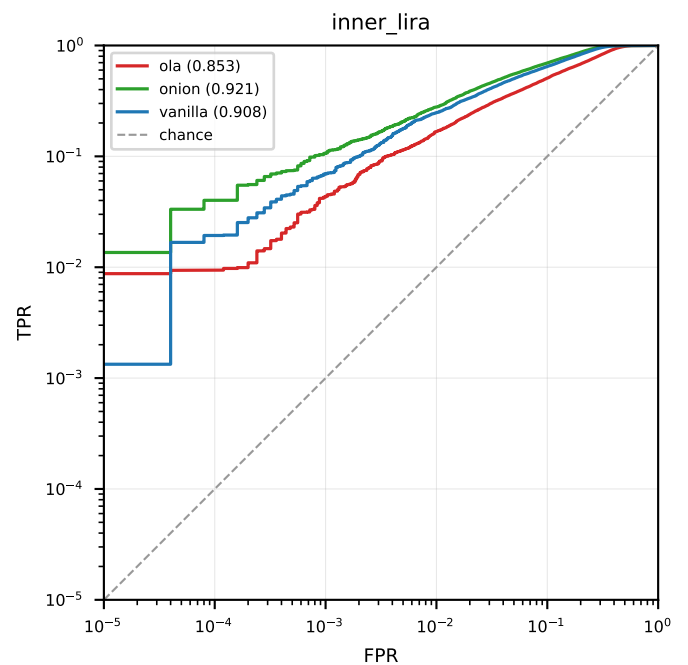
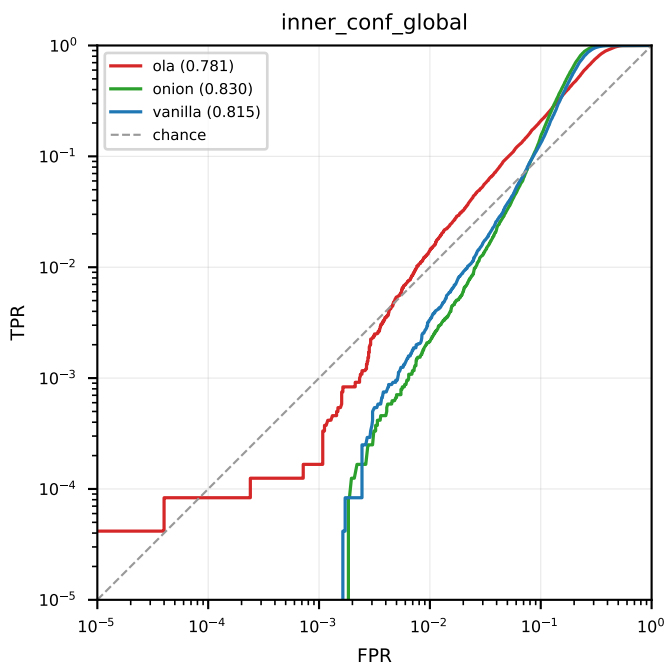
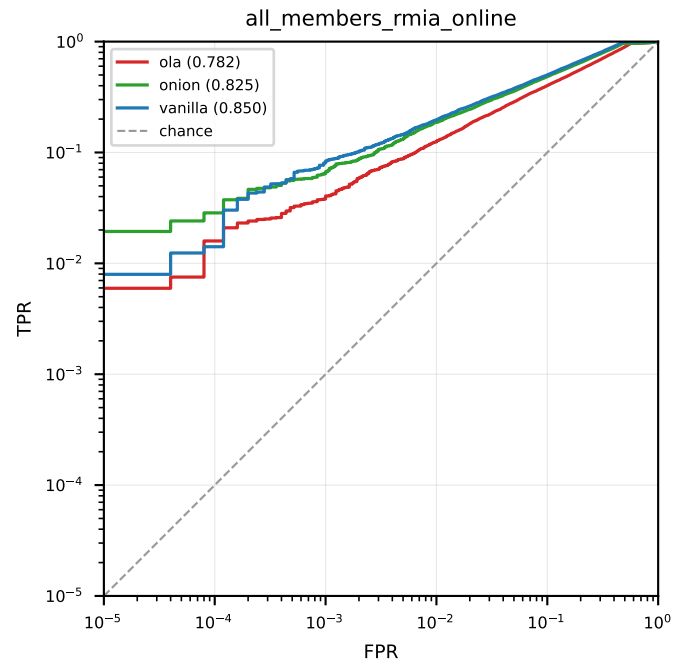
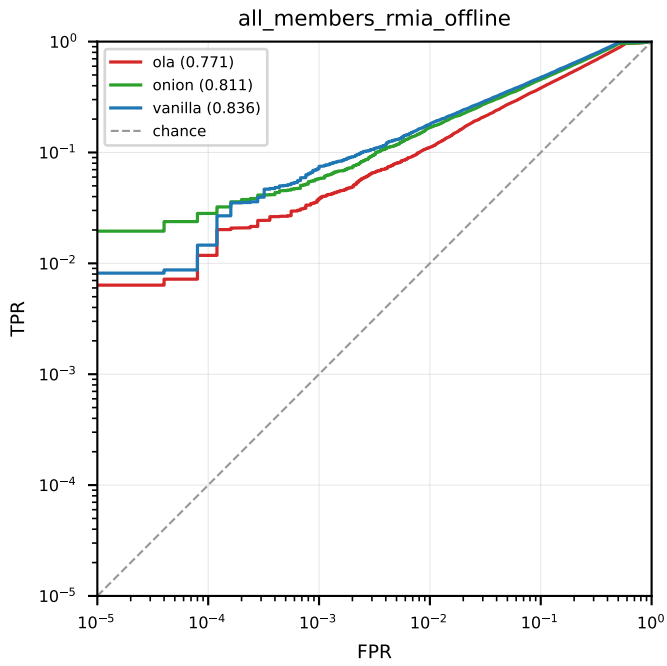
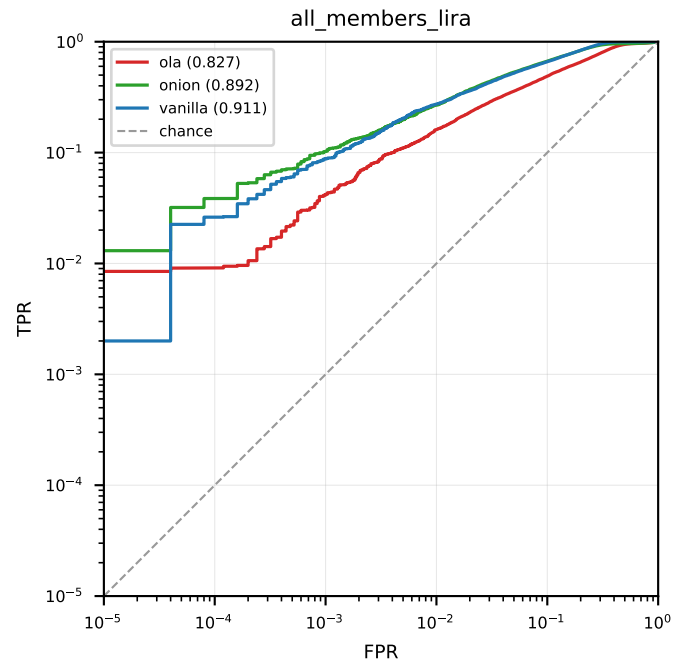
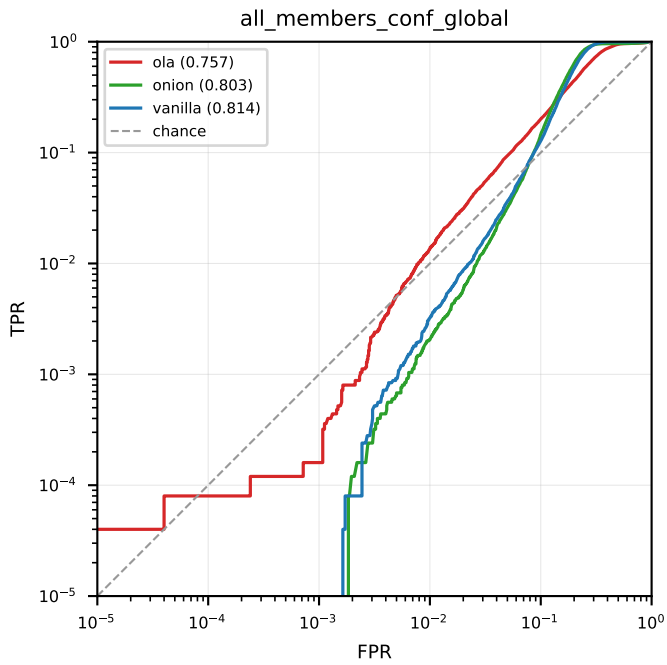
Positive control

dead_scorers : none
 weak_ceiling_scorers : none
 null_se : 0.00258
 chance_cut : 0.50516
 min_ceiling_auc : 0.55000
 vanilla AUC : rmia_offline=0.8359, rmia_online=0.8500, lira=0.9112, conf_global=0.8139

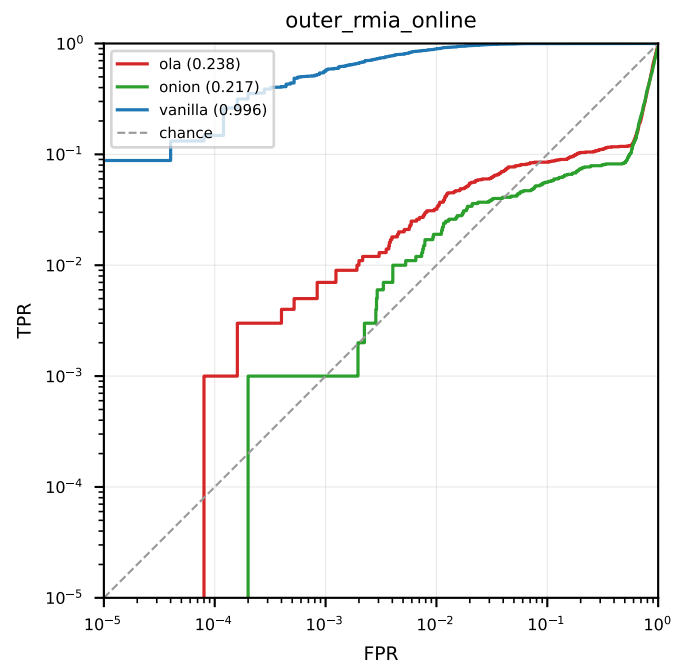
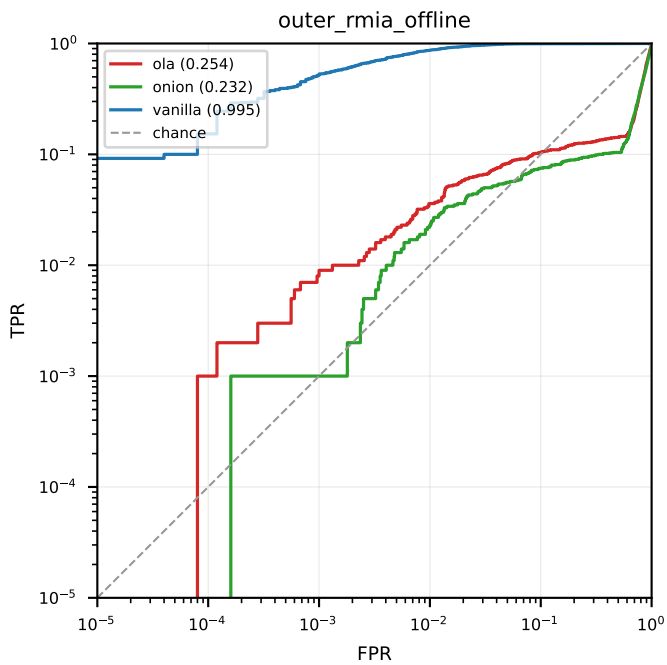
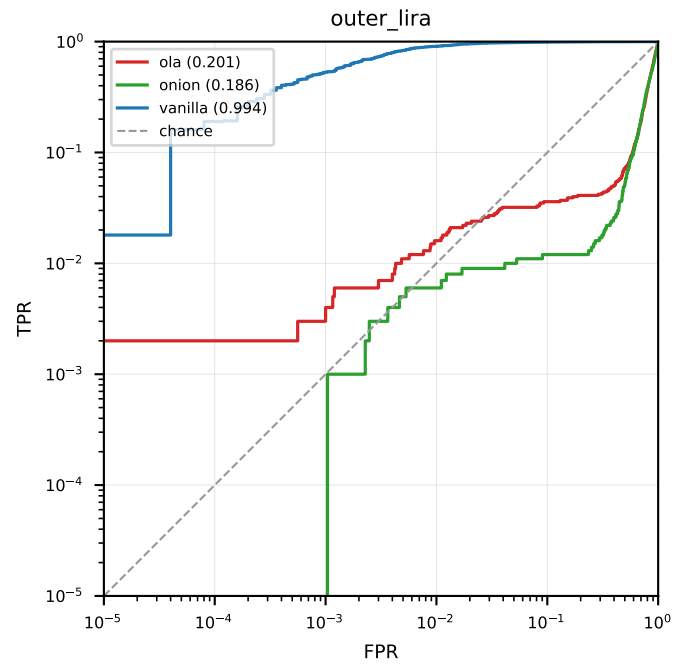
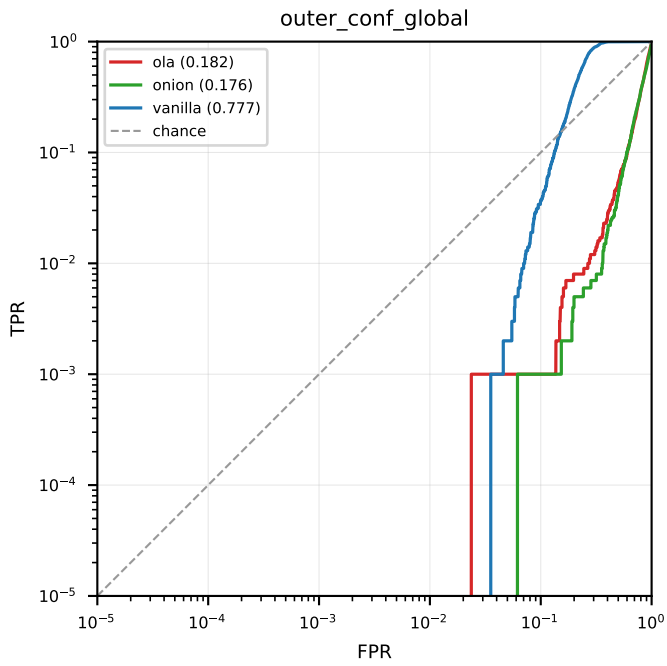
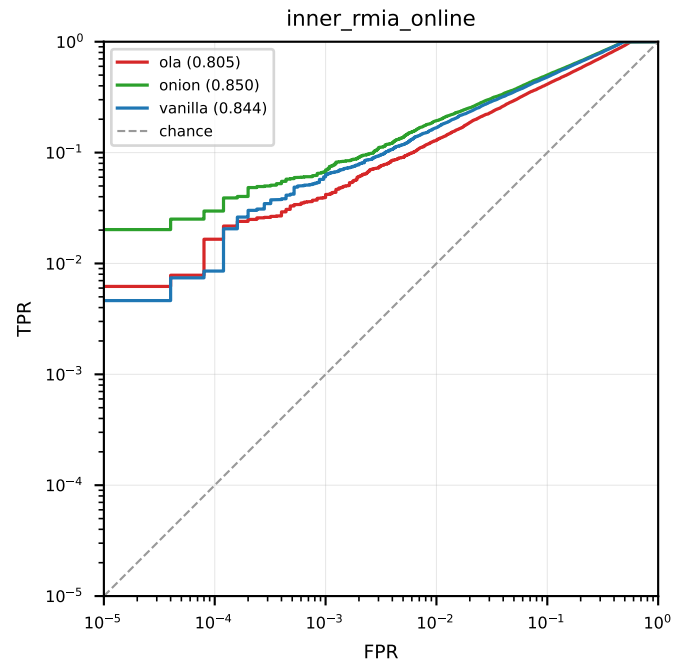
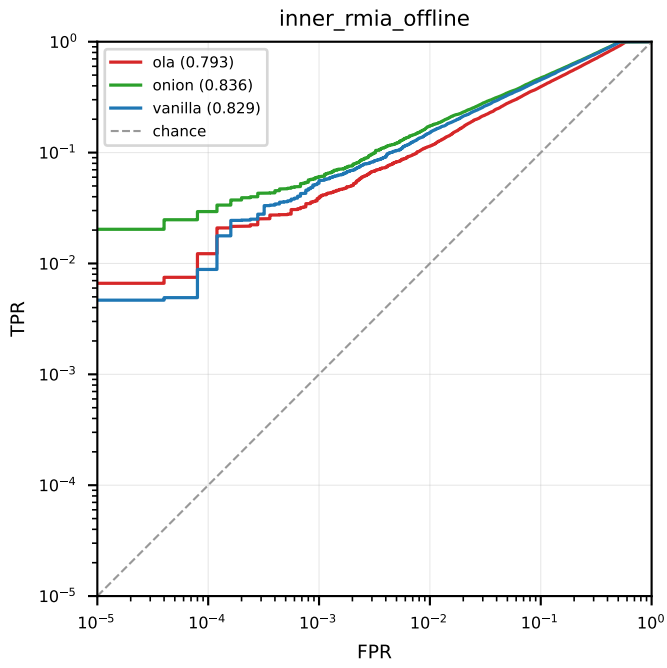
Timing

Training model 0 took 245.4543058872223 seconds
Training model 1 took 240.28910732269287 seconds
Training model 2 took 242.04000091552734 seconds
Training model 3 took 241.2706859111786 seconds
Training model 4 took 241.52451753616333 seconds
Training model 5 took 241.7392451763153 seconds
Training model 6 took 241.74009776115417 seconds
Training model 7 took 241.86516046524048 seconds
Training model 8 took 241.64260363578796 seconds
Training model 9 took 241.8999137878418 seconds
Training model 10 took 242.3395802974701 seconds
Training model 11 took 242.08797121047974 seconds
Training model 12 took 241.79558563232422 seconds
Training model 13 took 242.74364233016968 seconds
Training model 14 took 242.14570236206055 seconds
Training model 15 took 243.25878524780273 seconds
Training model 16 took 242.19645977020264 seconds
Training model 17 took 243.31837034225464 seconds
Training model 18 took 241.25903010368347 seconds
Training model 19 took 242.41969656944275 seconds
Training model 20 took 242.2129032611847 seconds
Training model 21 took 242.20262217521667 seconds
Training model 22 took 242.04069113731384 seconds
Training model 23 took 244.6887650489807 seconds
Training model 24 took 247.156968832016 seconds
Training model 25 took 243.65465354919434 seconds
Training model 26 took 242.019198179245 seconds
Training model 27 took 242.15695333480835 seconds
Training model 28 took 241.5683834552765 seconds
Training model 29 took 241.23206281661987 seconds
Training model 30 took 242.20028424263 seconds
Training model 31 took 243.97004556655884 seconds
Training model 32 took 242.15872430801392 seconds
Training model 33 took 245.29743361473083 seconds
Training model 34 took 247.51834225654602 seconds
Training model 35 took 241.69451332092285 seconds
Training model 36 took 242.59542727470398 seconds
Training model 37 took 242.60102462768555 seconds
Training model 38 took 249.45883131027222 seconds
Training model 39 took 242.46479606628418 seconds
Training model 40 took 242.71725583076477 seconds
Training model 41 took 244.75595879554749 seconds
Training model 42 took 250.46051740646362 seconds
Training model 43 took 248.83173036575317 seconds
Training model 44 took 247.650230884552 seconds
Training model 45 took 245.45027780532837 seconds
Training model 46 took 242.5113604068756 seconds
Training model 47 took 242.22493267059326 seconds
Training model 48 took 242.29372596740723 seconds
Training model 49 took 242.8914840221405 seconds
Training model 50 took 242.3744502067566 seconds
Training model 51 took 243.43781161308289 seconds
Training model 52 took 242.84735703468323 seconds
Training model 53 took 243.5500786304474 seconds
Training model 54 took 241.78812837600708 seconds
Training model 55 took 242.14992427825928 seconds
Training model 56 took 242.7299666404724 seconds
Training model 57 took 242.4132740497589 seconds
Training model 58 took 243.67469000816345 seconds
Training model 59 took 243.10644340515137 seconds
Training model 60 took 241.6673619747162 seconds
Training model 61 took 244.38061928749084 seconds
Training model 62 took 244.01380157470703 seconds
Training model 63 took 243.7456750869751 seconds
Shadow pool ready in 15601.7 s
Shadow signals took 45.9 s
Target models ready in 558.4 s
Signals took 31.3 s
Auditing took 16.0 s
Total runtime: 16255.5 s

sweep_ola_cifar10 — log-scale ROC



sweep_ola_cifar10 — log-scale ROC



sweep_ola_cifar100

OLA (Outer Layer Attenuation) · dataset cifar100

Configuration

outer_size : 1000
p_gen : 0.5
p_mem : 0.2
selection_fpr: 0.01
mask_seed : 12345
test_size : 5000

Accuracy

role	inner_acc	outer_acc	p_gen	p_mem	test_acc	train_size
vanilla	0.9122	0.9110	n/a	n/a	0.2392	25000
ola	0.3558	0.0410	0.5000	0.2000	0.1764	25000
onion	0.8704	0.0430	n/a	n/a	0.2388	24000

Subset: outer

AUC				
role	conf_global	lira	rmia_offline	rmia_online
vanilla	0.8865	0.9878	0.9484	0.9678
ola	0.2852	0.3403	0.3002	0.2883
onion	0.2280	0.2691	0.2370	0.2288

Two-sided AUC				
role	conf_global	lira	rmia_offline	rmia_online
vanilla	0.8865	0.9878	0.9484	0.9678
ola	0.7148	0.6597	0.6998	0.7117
onion	0.7720	0.7309	0.7630	0.7712

TPR @ 0.1% FPR				
role	conf_global	lira	rmia_offline	rmia_online
vanilla	0.0000	0.5340	0.2700	0.1780
ola	0.0000	0.0030	0.0000	0.0000
onion	0.0000	0.0030	0.0000	0.0000

Subset: inner

AUC				
role	conf_global	lira	rmia_offline	rmia_online
vanilla	0.8913	0.9584	0.9324	0.9576
ola	0.6956	0.7100	0.7091	0.7232
onion	0.8748	0.9520	0.9133	0.9436

Two-sided AUC				
role	conf_global	lira	rmia_offline	rmia_online
vanilla	0.8913	0.9584	0.9324	0.9576
ola	0.6956	0.7100	0.7091	0.7232
onion	0.8748	0.9520	0.9133	0.9436

TPR @ 0.1% FPR				
role	conf_global	lira	rmia_offline	rmia_online
vanilla	0.0003	0.1208	0.0830	0.1161
ola	0.0012	0.0163	0.0139	0.0105
onion	0.0002	0.1612	0.0734	0.1152

Subset: all_members

AUC				
role	conf_global	lira	rmia_offline	rmia_online
vanilla	0.8911	0.9595	0.9330	0.9580
ola	0.6792	0.6952	0.6927	0.7058
onion	0.8489	0.9247	0.8862	0.9151

Two-sided AUC

role	conf_global	lira	rmia_offline	rmia_online
vanilla	0.8911	0.9595	0.9330	0.9580
ola	0.6792	0.6952	0.6927	0.7058
onion	0.8489	0.9247	0.8862	0.9151

TPR @ 0.1% FPR

role	conf_global	lira	rmia_offline	rmia_online
vanilla	0.0003	0.1373	0.0905	0.1186
ola	0.0012	0.0158	0.0133	0.0101
onion	0.0002	0.1549	0.0704	0.1106

Excess over the onion floor (two-sided AUC difference)

outer (null SE 0.00931)

	conf_global	lira	rmia_offline	rmia_online
excess	-0.0571	-0.0712	-0.0632	-0.0595
in null SE	-6.1344	-7.6482	-6.7900	-6.3934

inner (null SE 0.00261)

	conf_global	lira	rmia_offline	rmia_online
excess	-0.1792	-0.2420	-0.2042	-0.2205
in null SE	-68.6852	-92.7762	-78.2844	-84.5039

all_members (null SE 0.00258)

	conf_global	lira	rmia_offline	rmia_online
excess	-0.1697	-0.2295	-0.1935	-0.2093
in null SE	-65.7364	-88.8851	-74.9526	-81.0424

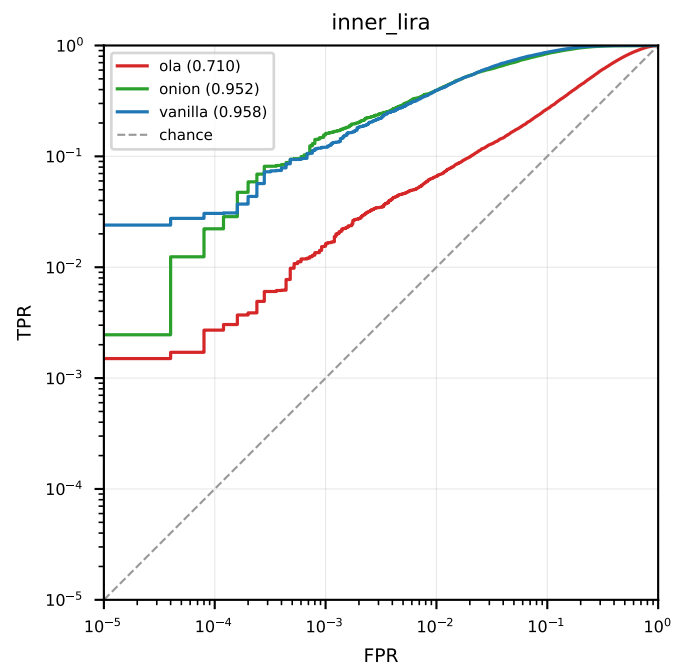
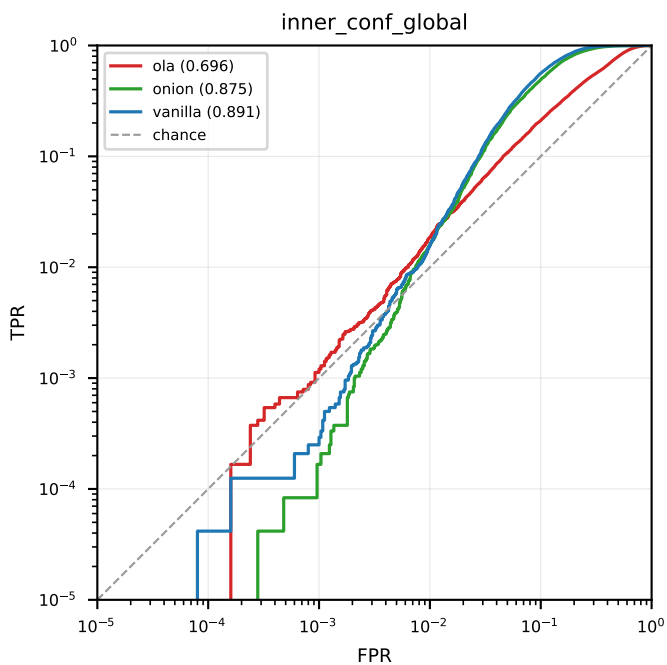
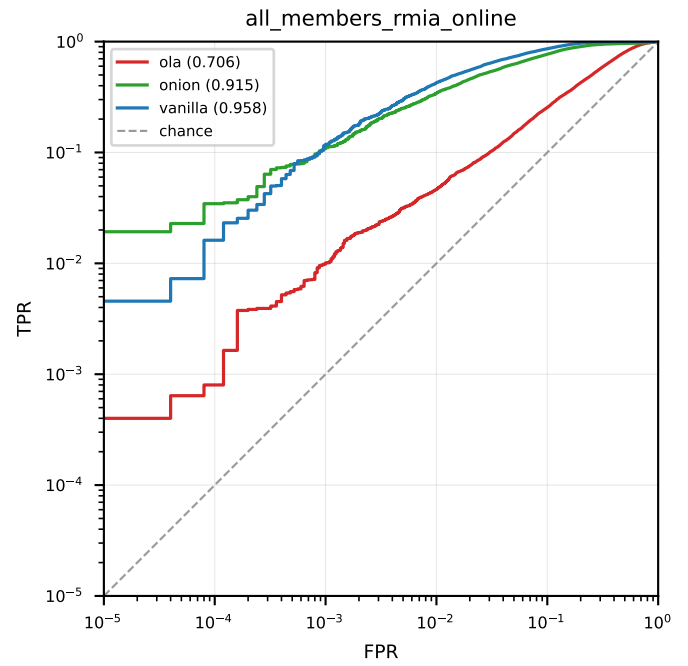
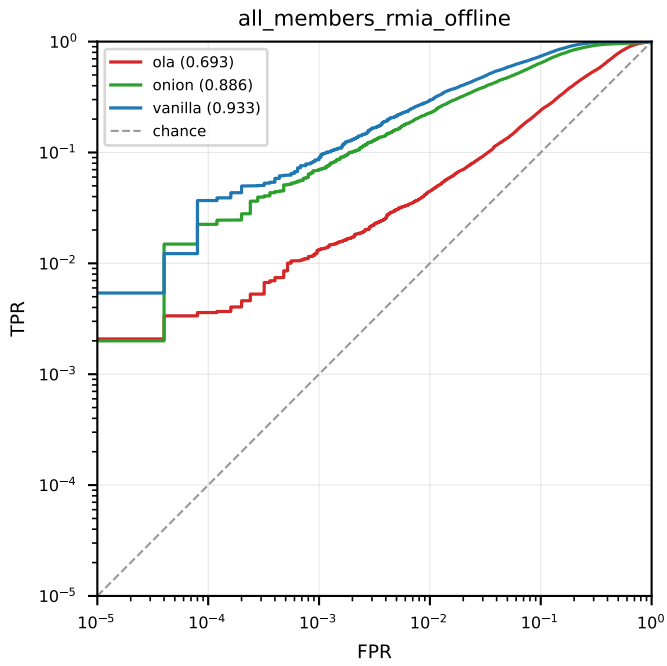
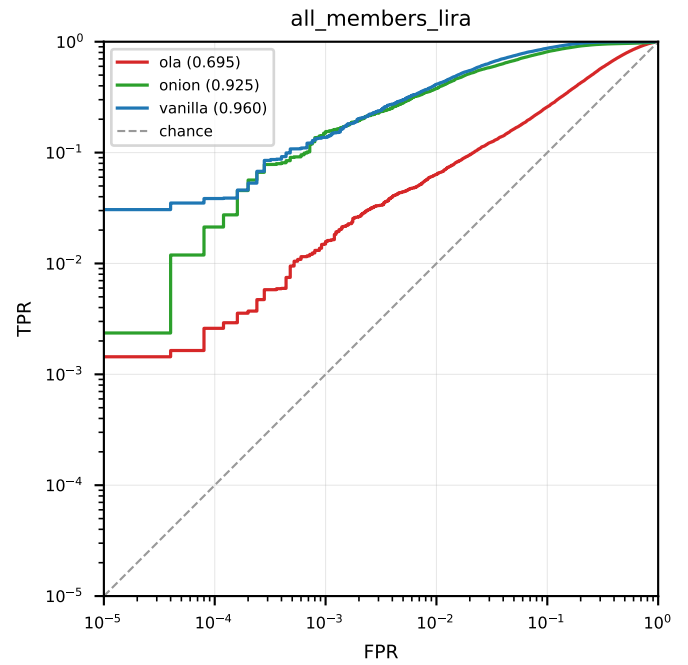
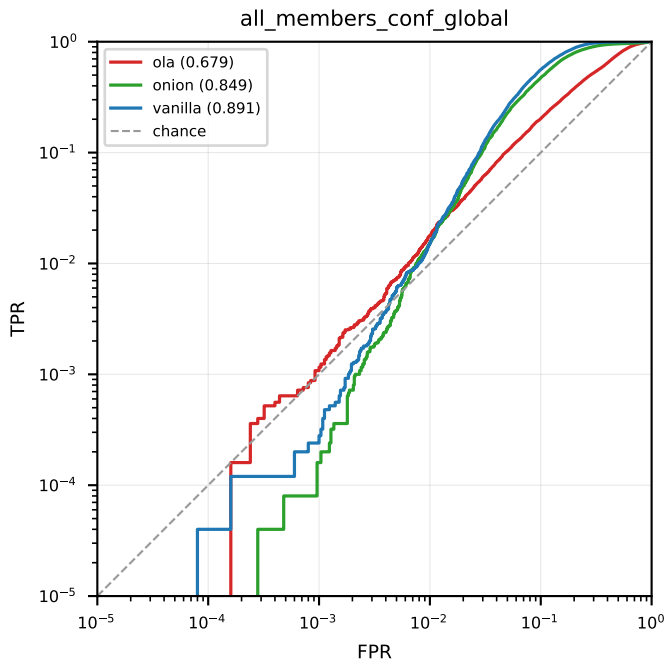
Positive control

dead_scorers : none
 weak_ceiling_scorers : none
 null_se : 0.00258
 chance_cut : 0.50516
 min_ceiling_auc : 0.55000
 vanilla AUC : rmia_offline=0.9330, rmia_online=0.9580, lira=0.9595, conf_global=0.8911

Timing

Training model 0 took 245.21882843971252 seconds
Training model 1 took 241.67260313034058 seconds
Training model 2 took 240.00573897361755 seconds
Training model 3 took 241.42771983146667 seconds
Training model 4 took 242.1970992088318 seconds
Training model 5 took 242.34537482261658 seconds
Training model 6 took 240.6487193107605 seconds
Training model 7 took 242.09291076660156 seconds
Training model 8 took 241.99281883239746 seconds
Training model 9 took 242.02532243728638 seconds
Training model 10 took 241.20100498199463 seconds
Training model 11 took 241.28466272354126 seconds
Training model 12 took 240.2742199897766 seconds
Training model 13 took 242.3152527809143 seconds
Training model 14 took 241.00831723213196 seconds
Training model 15 took 241.0959198474884 seconds
Training model 16 took 241.37981033325195 seconds
Training model 17 took 241.70721244812012 seconds
Training model 18 took 241.42171382904053 seconds
Training model 19 took 241.23594570159912 seconds
Training model 20 took 241.23970890045166 seconds
Training model 21 took 241.9213948249817 seconds
Training model 22 took 240.93606805801392 seconds
Training model 23 took 241.91986656188965 seconds
Training model 24 took 241.56590175628662 seconds
Training model 25 took 241.83772683143616 seconds
Training model 26 took 241.43828225135803 seconds
Training model 27 took 242.07203722000122 seconds
Training model 28 took 242.35660433769226 seconds
Training model 29 took 242.17837166786194 seconds
Training model 30 took 241.8276081085205 seconds
Training model 31 took 241.367116689682 seconds
Training model 32 took 241.81524634361267 seconds
Training model 33 took 241.88849592208862 seconds
Training model 34 took 241.72133827209473 seconds
Training model 35 took 241.78275680541992 seconds
Training model 36 took 241.84224319458008 seconds
Training model 37 took 242.90130639076233 seconds
Training model 38 took 241.65316247940063 seconds
Training model 39 took 241.88628482818604 seconds
Training model 40 took 240.49641609191895 seconds
Training model 41 took 242.2451617717743 seconds
Training model 42 took 241.9260392189026 seconds
Training model 43 took 240.8048951625824 seconds
Training model 44 took 242.03784680366516 seconds
Training model 45 took 241.46363878250122 seconds
Training model 46 took 241.9826295375824 seconds
Training model 47 took 241.22024059295654 seconds
Training model 48 took 242.4492084980011 seconds
Training model 49 took 241.76564860343933 seconds
Training model 50 took 242.53996348381042 seconds
Training model 51 took 241.9994194507599 seconds
Training model 52 took 242.1670105457306 seconds
Training model 53 took 242.29312300682068 seconds
Training model 54 took 242.27486777305603 seconds
Training model 55 took 243.05627918243408 seconds
Training model 56 took 242.2268967628479 seconds
Training model 57 took 242.04895448684692 seconds
Training model 58 took 242.80448579788208 seconds
Training model 59 took 243.027006149292 seconds
Training model 60 took 242.73222374916077 seconds
Training model 61 took 242.55367708206177 seconds
Training model 62 took 241.81476950645447 seconds
Training model 63 took 242.87246322631836 seconds
Shadow pool ready in 15520.0 s
Shadow signals took 45.6 s
Target models ready in 555.3 s
Signals took 30.6 s
Auditing took 16.1 s
Total runtime: 16170.1 s

sweep_ola_cifar100 — log-scale ROC



sweep_ola_cifar100 — log-scale ROC

